Please do not remove this page

Multi-objective optimization of bread waste valorization: a framework for integrated food waste management

Hafyan, Rendra Hakim; Sadhukhan, Jhuma; Gadkari, Siddharth

https://openresearch.surrey.ac.uk/esploro/outputs/journalArticle/Multi-objective-optimization-of-bread-waste-valorization/991103995002346/filesAndLinks?index=0

Hafyan, R. H., Sadhukhan, J., & Gadkari, S. (2026). Multi-objective optimization of bread waste valorization: a framework for integrated food waste management. Resources, Conservation and Recycling, 229(), Article 108871. https://doi.org/10.1016/j.resconrec.2026.108871
Document Version: Author's Accepted Manuscript

Published Version: https://doi.org/10.1016/j.resconrec.2026.108871

Surrey Open Research repository homepage: https://openresearch.surrey.ac.uk/esploro/
openresearch@surrey.ac.uk
CC BY V4.0
Local
Downloaded On 2026/03/26 21:15:36 +0000

Multi-Objective Optimization of Bread Waste Valorization: A Framework for Integrated Food Waste Management

Rendra Hakim Hafyan[a], Jhuma Sadhukhan[b,c], Siddharth Gadkari[a,b*]

[a] *School of Chemistry and Chemical Engineering, University of Surrey, Guildford GU2 7XH, UK*
[b] *Institute for Sustainability, University of Surrey, Guildford GU2 7XH, UK*
[c] *Centre for Environment & Sustainability (CES), School of Sustainability, Civil and Environmental Engineering, University of Surrey, Guildford GU2 7XH, UK*

ABSTRACT

Bread waste is a major contributor to food waste in the UK, creating significant environmental and resource management challenges. This study develops an integrated bread waste biorefinery framework to support circular bioeconomy objectives by simultaneously addressing environmental, economic, and social performance. A multi-objective optimisation model, formulated as a mixed-integer nonlinear programming problem and solved using an augmented ε-constraint approach, was applied to optimise net present value (NPV), global warming potential (GWP) savings, and employment generation. The framework integrates anaerobic digestion and incineration with biorefinery pathways under three heat supply strategies: fully internal, partially external, and fully external heat provision. Results reveal clear trade-offs between objectives. Fully internal heat supply maximises environmental performance, achieving up to 361,741 tCO_2-eq yr^{-1} GWP savings, while external heat supply delivers superior economic and social outcomes, with NPV up to 733 M$ and 4,881 jobs. The proposed framework provides robust decision support for sustainable food-waste management.

Keywords: Bread waste valorization; Integrated biorefinery; Multi-objective optimization; Food waste management; Circular bioeconomy

Nomenclature

Abbreviations

BW	Bread waste
FW	Food waste
IBWR	Integrated bread waste biorefinery
MT	Metric tonne
M$	Million dollar
SA	Succinic acid
LA	Lactic acid
MOO	Multi-objective optimization
LCA	Life cycle assessment
MINLP	Mixed-integer non-linear programming
MILP	Mixed-integer linear programming
AD	Anaerobic digestion
IFBR	Integrated forest biorefineries
BDO	2,3-butanediol
CCU	Carbon capture unit
GGSS	Green Gas Support Scheme
SDG	Sustainable Development Goals
SW	Solid waste
GIS	Geographic information system
MCS	Monte Carlo simulation

1. Introduction

The rapid rise in population, industrialization, and food consumption has increased pressure on food supplies, land, and fossil fuels, leading to environmental issues like climate change and resource depletion. With declining fossil fuel reserves and rising waste, the shift to renewable and sustainable alternatives is urgently needed (Nikolaou et al., 2023; Ouro-Salim & Guarnieri, 2022). BW represents a valuable resource for the production of bioenergy and biochemicals due to its rich composition, which typically comprises 50–70% starch, 8–10% protein, and 1–5% lipids by weight (Kumar et al., 2023). Its high carbohydrate content makes it particularly suitable for generating glucose-based products (Hafyan et al., 2024a). In addition, its lipids can be converted into biodiesel, while proteins serve the food, medical, and cosmetic industries, supporting a broad biorefinery approach (Karmee et al., 2015; Kumar et al., 2023; Svensson et al., 2021). Notably, bread is the UK's second most wasted food, with 292,000 tons discarded annually, generating 584,000 tons of CO_2-equivalent emissions, underscoring its significant environmental and economic impact (Narisetty et al., 2021). BW occurs throughout the supply chain due to inadequate transport, storage, handling inefficiencies, and consumer habits such as poor purchasing improper storage (Jeswani et al., 2021; Sinha & Tripathi, 2021). Unlike lignocellulosic biomass, which requires intensive processing, BW provides a clean, easily extractable source of fermentable sugars (Hafyan et al., 2024a; Kumar et al., 2023). This makes it a promising feedstock for supporting a low-carbon and circular bioeconomy (Brandão et al., 2021; Yaashikaa et al., 2020).

In the UK, a waste management hierarchy prioritizes waste prevention, redistribution, animal feed, composting, AD, and incineration, with landfill as a last resort(Facchini et al., 2018). While various studies have demonstrated the potential of BW valorization, such as the production of bioethanol and succinic acid (SA) (Hafyan et al., 2024b), LA and biomethane (Mailaram et al., 2023), hydrogen (Hafyan et al., 2025), and ethanol via fermentation (Narisetty et al., 2022), most have focused on the economic and environmental assessments of isolated processes rather than integrated system optimization. For example, Hafyan et al. (2024b) found that combining SA and bioethanol production, carbon capture, and energy generation could yield both economic and environmental benefits. Other studies have highlighted the viability and profitability of lactic acid (LA) production under specific conditions (Mailaram et al., 2023), or the comparable emissions profile of BW-derived bioethanol relative to conventional feedstocks (Narisetty et al., 2022). However, holistic approaches that optimize across multiple objectives and integrate biorefinery with FW management remain limited.

Designing a biorefinery for waste valorization requires algorithms to evaluate process conversion pathways based on performance criteria (Sadhukhan et al., 2014). However, challenges arise due to the diverse feedstocks, products, and processing methods (Maity, 2015). A systematic approach is needed to integrate processes while considering feedstock availability and market demands. In this context, multi-objective optimization (MOO) plays a crucial role in balancing economic viability and environmental sustainability (Thomassen et al., 2019), helping identify optimal configurations that enhance the efficiency and sustainability of biorefinery operations (Haghpanah et al., 2022). Recent studies have designed biorefinery superstructures that can handle multiple feedstocks and produce a variety of products. Galanopoulos et al. (2019) developed a framework to optimize an integrated algae biorefinery, reducing biodiesel production costs by up to 80% compared to standalone biorefineries. Their study used wastewater and CO_2 emissions from a wheat straw biorefinery as feedstock, targeting products like biodiesel, glycerol, bioethanol, and levulinic acid. Similarly, Ben Daya and Nourelfath (2019) developed a MOO model to enhance sustainability in Integrated Forest Biorefineries (IFBR) at Canadian pulp mills, using feedstocks like forest residues, agricultural waste, and municipal solid waste. Their approach, combining MOO and life cycle assessment (LCA), aimed to reduce emissions,

maximize financial value, and enable pulp mills to diversify into clean energy production while improving economic viability. Wu et al. (2024) optimized lignin valorization pathways in biorefineries using a MINLP model. They focused on feedstocks such as hardwood, softwood, and herbaceous plants to produce products like 2-pyrone-4,6-dicarboxylic acid, polyhydroxyalkanoates, biodiesel, and adipic acid. The study found that producing PDC from hardwood was the most economically promising option, with an estimated NPV of 771.41 M$. Giuliano et al. (2016) optimized a multi-product biorefinery by considering seasonal variations in lignocellulosic biomass. Using mixed-integer linear programming (MILP), they analyzed biomass allocation and process routes with feedstocks like eucalyptus residues, wheat straw, and olive tree prunings, targeting LA, SA, and ethanol. The study found that biomass composition significantly influenced product allocation and process performance, with the optimal flowsheet varying depending on feedstock seasonality.

An analogous approach can be applied to waste management in the UK, where the government aims to reduce biodegradable municipal waste sent to landfills and contribute to Sustainable Development Goals (SDG) 12.3 and 12.4, specifically halving per capita global FW by 2030 and ensuring the environmentally sound management of chemicals and all wastes throughout their life cycle (Frank et al., 2017; Slorach et al., 2020). This would enhance food security, reduce environmental impact, and support the Global Methane Pledge commitments (Bakkaloglu et al., 2022). While AD and incineration are common practices to divert waste from landfills, a more strategic and integrated approach is necessary to meet the UK's 2050 net-zero emissions target and support the transition to a circular economy. Combining FW management with advanced biorefinery processes presents a promising solution for efficiently converting waste into valuable products; this approach is also effective in reducing the amount of FW sent to landfills. To the best of our knowledge, no prior studies have explored the integration of BW valorization into FW management practices using MOO. Also, while current FW management strategies, such as AD and incineration, are well-established, the opportunity to further improve sustainability through the incorporation of biorefinery technologies has yet to be fully explored. We present a novel superstructure-based model that integrates traditional FW management (e.g., AD and incineration) with advanced biorefinery processes (e.g., fermentation, lipid upgrading, protein extraction, and carbon capture). A MINLP approach is employed to simultaneously optimize economic (NPV), environmental (GWP savings), and social (employability) objectives. A decision-making framework for stakeholders identifies optimal solutions, offering diverse options for efficient, sustainable, and socially beneficial FW management.

This study proposes the integrated bread waste biorefinery (IBWR) model as a framework to optimize the valorization of BW within the existing FW management hierarchy. The key challenge lies in evaluating the sustainability of this integration, guided by three primary objectives: maximizing the NPV of the IBWR, maximizing GWP savings, and maximizing employability. In addition, the selection of optimal processing routes is essential for achieving overall sustainability, as different configurations of the biorefinery system affect the economic, environmental, and social performance of the IBWR. This highlights the need for optimization models to help decision-makers in developing efficient and sustainable solutions. The study explores biorefinery pathways for converting BW into valuable products:

- **Incineration pathway**: Produces electricity and heat from BW and biogas through cogeneration technology.
- **Anaerobic digestion pathway**: Utilizes AD to generate biogas from BW, which can be further purified through biogas upgrading using membrane technology to obtain pure biomethane.
- **Biorefinery pathway**: This pathway consists of three main routes: the Fermentation pathway, which produces carbohydrate-based products such as bioethanol, 2,3-butanediol (2,3-BDO), SA, butanol and LA through fermentation; the Transesterification pathway, which converts lipids from

BW into biodiesel via transesterification; and the Bioactive Compound pathway, which extracts proteins from BW to produce bioactive compounds. These predominant products of biorefineries have been selected due to all contributing significantly to the global bio-based market (Sadhukhan et al., 2019; Sadhukhan & Martinez-Hernandez, 2017; Sadhukhan et al., 2024; Sadhukhan et al., 2016).

- **Carbon capture unit (CCU) pathway**: CCU involves capturing or separating CO_2 from flue gas, transporting it to a storage site, and utilizing a portion of it for SA production.

In addition to these objectives, the study seeks to answer the following two critical questions below:

- **Which investment strategies and process configurations will optimize the economic, environmental, and social outcomes of the IBWR system?** Specifically, which biorefinery pathways and technologies should be prioritized in each phase of investment, and what are the optimal production capacities to implement?
- **How can the integration of BW valorization processes within the IBWR system be managed to achieve the most effective synergy?** This question explores the coordination of FW management practices (such as AD and incineration) with biorefinery technologies to ensure an optimal balance of energy generation, economic performance, waste minimization, and job creation.

2. Methodology

FW typically contains three major macronutrients (carbohydrates, proteins, and lipids), which can serve as precursors for producing biofuels and biochemicals. In conventional FW management, these components are mainly utilized through processes such as AD and incineration, primarily for energy and biogas recovery. While the biochemical composition of FW suggests that advanced biorefinery routes could offer additional pathways for value recovery and improved sustainability, the extent of their economic, environmental, and social benefits remains insufficiently quantified and systematically optimized. To address this gap, a generic Integrated Food Waste Valorization Framework is proposed to enable the structured evaluation and optimization of FW management by integrating established practices (e.g., AD and incineration) with emerging biorefinery technologies. The framework is designed to be adaptable to different FW streams by parameterizing their compositional characteristics (carbohydrates, lipids, and proteins), allowing the identification of feasible conversion pathways and optimal process configurations. As a case study to demonstrate the application of this generalized framework, BW is selected due to its high carbohydrate and protein content and its status as the second most wasted food category in the UK.

The methodology for optimizing the IBWR system is structured into four key steps that align with concept definition, process simulation, optimization, and data analysis, as depicted in Figure 1. This approach integrates BW valorization and FW management, optimizing for economic, environmental, and social objectives. Below is an elaboration based on the four steps and their combination with the previously provided methodology description:

- **Step 1 (Concept Definition):** The first step defines the concept of BW utilization. BW can follow two pathways: one focusing on FW management through recycling (AD) and recovery (incineration for energy production), and the other on valorization to produce valuable biochemicals (e.g., 2,3-BDO, LA, and SA), biofuels (e.g., bioethanol, butanol, biodiesel) and bioactive compound (e.g., protein). Both pathways can be integrated into the IBWR system, where FW management and BW valorization work synergistically. This superstructure represents all possible process alternatives and serves as the foundation for the subsequent steps.
- **Step 2 (Process Simulation):** The next step involves simulating all processes using Aspen Plus v11. In this step, critical data, such as mass and energy balances, are generated for both pathways of

the IBWR system. Appropriate thermodynamic models for liquid and vapor phases are selected to ensure accuracy in simulation. The data extracted from the simulations, mass flow rates, energy consumption, and byproduct generation, are then used to inform the optimization model.

- **Step 3 (Optimization):** After process simulation data is obtained, the optimization phase begins. Mass and energy balances are used to construct the MOO model in Python, considering three objectives: maximizing NPV (economic), GWP savings (environmental), and employability (social). A MINLP model is formulated to determine the most sustainable IBWR configuration, balancing all objectives.
- **Step 4 (Data Analysis):** The final step involves analyzing the results of the optimization using Pareto front analysis. The Pareto front allows for visualizing the trade-offs between economic, environmental, and social objectives. By examining the Pareto-optimal solutions, decision-makers can evaluate the compromises required to optimize the IBWR system holistically. This analysis is crucial for determining the best process configuration that aligns with sustainable development goals.

Overall, this methodology integrates BW valorization and FW management processes, with an emphasis on optimizing the system using multi-objective criteria. The complete list of indices, sets, parameters, and decision variables necessary for constructing the optimization model is detailed in Table A1 of the supplementary material, providing a clear framework for reproducibility and further refinement of the IBWR system. In addition, the yields of glucose-based products, validated against the references used in this study, are summarized in Table A4.

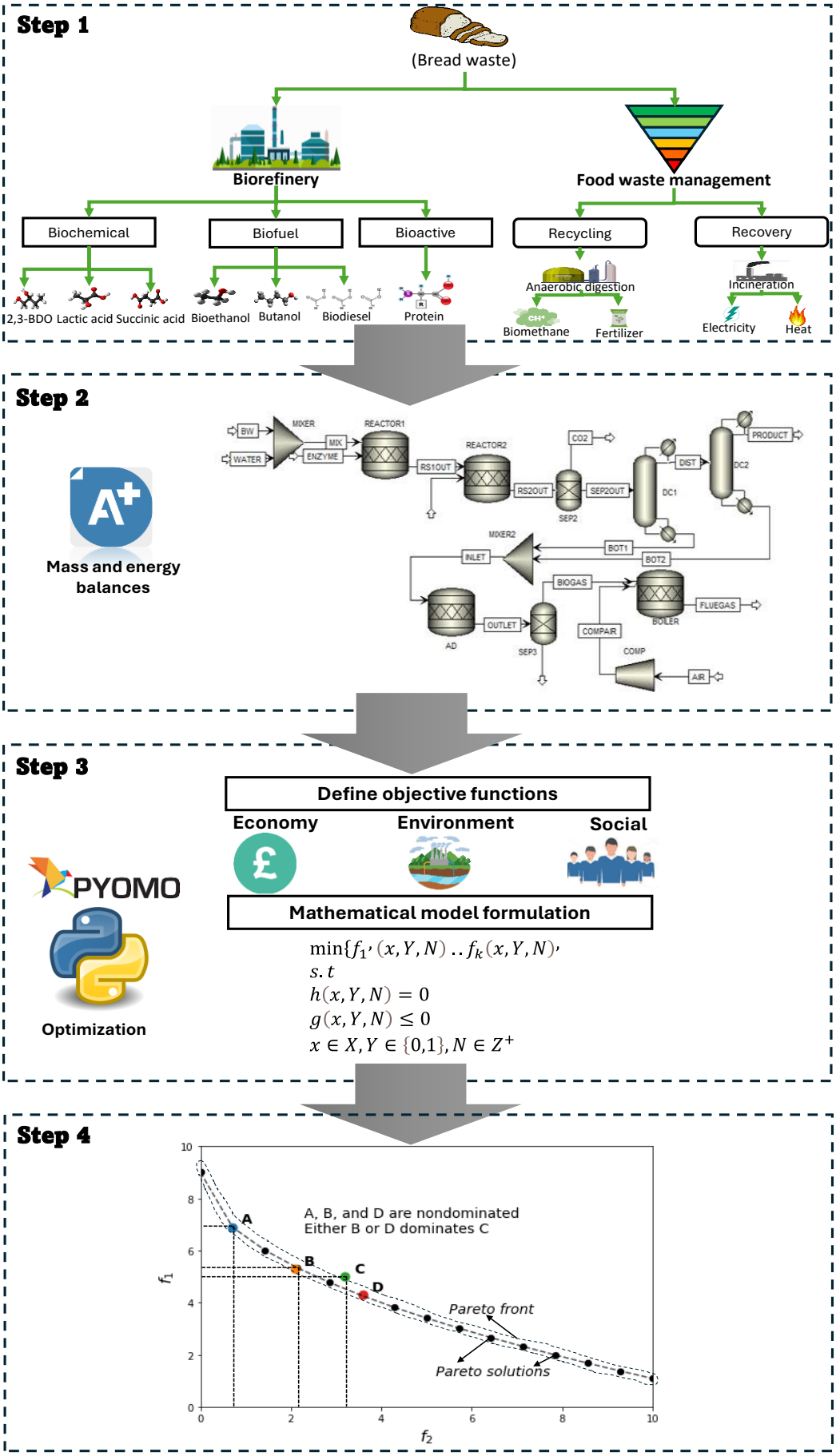

Fig.1. The flowchart of the methodology for optimizing the IBWR system

2.1. Superstructure

The IBWR superstructure model represents the specific implementation of the proposed framework. It optimizes BW valorization by integrating FW management methods like AD and incineration with advanced biorefinery technologies. By efficiently managing mass and energy flows, as described in Figure 2, the model ensures that BW is converted into high-value products, including biofuels, chemicals, and energy. Each process stage offers multiple technological alternatives, selecting the best configuration based on economic, environmental, social and technical criteria. Comprehensive modeling maximizes resource use, minimizes waste, and ensures energy efficiency while producing sustainable, economically viable bioproducts. The superstructure represents a network of interconnected processes, emphasizing the efficient utilization of BW. The composition of BW used in the model is as follows: 46 wt% starch, 7.9 wt% protein, 2 wt% fat, 2.5 wt% fiber, 1 wt% salt, and 40.6 wt% water (Narisetty et al., 2022). This composition informs subsequent conversion pathways, ensuring the system fully exploits BW's potential for valuable products and energy. The IBWR model thus exemplifies an integrated strategy for FW management transforming BW into a high-value biorefinery feedstock and demonstrating how the proposed framework can be extended to other FW types with similar biochemical compositions. The superstructure model focuses on mass and energy integration, effectively using all materials and energy streams. BW is fractionated into carbohydrates, lipids, and proteins, each following distinct pathways. Carbohydrates are converted to glucose for fermentation into chemicals and biofuels, lipids are processed into biodiesel, and proteins are purified for specialized uses, maximizing the recovery of valuable bioproducts.

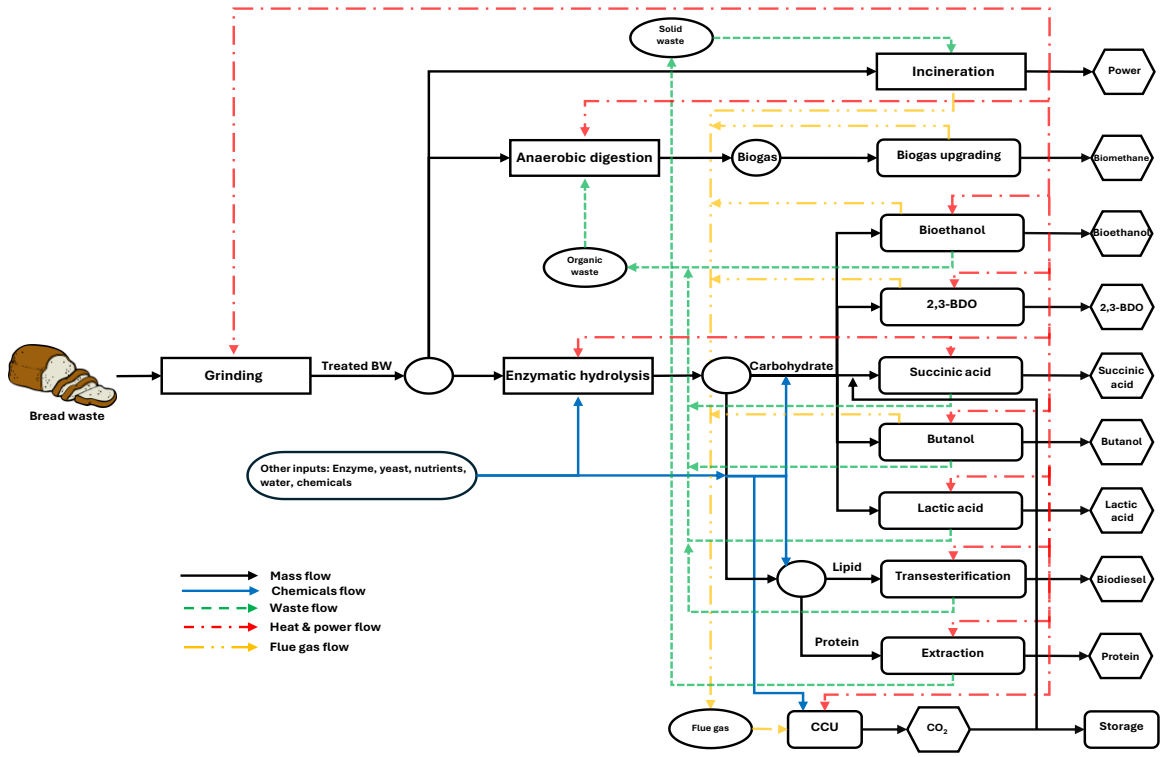

Fig.2. The developed superstructure of IBWR system

Energy integration is essential to the IBWR's goal of maximizing resource recovery. AD and incineration are incorporated into the superstructure for waste treatment and energy generation, ensuring efficient resource utilization within the FW management hierarchy. For the upgrading of biogas to biomethane, a membrane technology for grid injection has been modelled. Incineration, another key component of traditional FW management, is utilized to combust BW and remaining solid

impurities to generate energy (heat and power). This energy fulfils the system's power demands for various processes involved in the IBWR system. The inclusion of incineration highlights its value in managing the energy needs of the biorefinery, ensuring that waste is not only minimized for external energy dependence but also transformed into useful energy. To promote the use of renewable energy sources in the UK, the government offers incentive scheme for AD and incineration. The purified biomethane obtained in the AD qualifies for the Green Gas Support Scheme (GGSS) (GOV.UK, 2024), a financial incentive in the UK for injecting biomethane into the grid, with tariff rates shown in Table A2 of the supplementary material. The GGSS offers biomethane injected into the grid, this is expected to increase the share of facilities upgrading biogas. Another incentive scheme applied is the average gate fee paid by local authorities to AD and incineration facilities (WRAP, 2024a).

The CCU plays a crucial role in the system's environmental performance. CO_2 generated during incineration and other fermentation processes contributes to GWP savings through energy or material recovery, thereby enhancing the overall sustainability of the process. In the case of CCU integration, captured CO_2 is converted into SA based on the molar ratio of glucose consumed. The required amount of CO_2 is determined by the stoichiometric relationship, where approximately one mole of CO_2 is incorporated into SA for every mole of glucose (Gadkari et al., 2021). Any excess CO_2 that is not consumed by the fermentation process is diverted into a CO_2 grid (a network for CO_2 transport) via pipelines for storage. Commercial CO_2 pipelines are designed to operate within pressure ranges of 86 to 152 bar and temperatures of 13°C to 43°C to maintain CO_2 in a dense state, which reduces transport costs and minimizes two-phase flow issues (Hong, 2022). These specifications protect pipeline integrity by avoiding coating damage at high temperatures and preventing issues with metal performance at low temperatures. To prepare CO_2 for transport and storage, it is compressed to approximately 150 bar, often in a liquid phase to reduce pressure requirements. In this study, a CO_2 compression process is simulated based on the work of Park et al. (2024) employing a multi-stage compression system with intercooling. The process involves initial gas compression, cooling to 40°C, and condensation in a flash drum, with only the remaining gas stream undergoing further compression.

The incorporation of energy-generating processes like AD and incineration allows the system to be self-sustaining, while CCU and reuse strategies contribute to its environmental performance. This holistic approach not only advances the sustainability of FW management but also enhances the economic and technical viability of the biorefinery system. Table A3 describes the technologies considered in this work. The model presented in this work addresses the strategic design of IBWR system for the production of biochemical and biofuel products considering multiple objectives.

2.2. Mathematical modelling

2.2.1. Economic objective

A techno-economic model is developed to estimate the total capital investment, manufacturing cost and revenue of the optimal processing routes for IBWR system. The assumptions in economic model are tabulated in Table A6 of the supplementary material. The years 1 to 3 of the project are meant for the plant construction, and the plant start-up is fixed at the beginning of year 4. The first objective function considered here is the NPV of the IBWR system. This imposes the maximization of profit-related indexes. The following economic objective function is to maximize the NPV of the IBWR system as expressed in Eq. 1.

$$Max\ Z_1\ (NPV) = \sum_{t=t_0}^{T} \left(\frac{dCCF_t}{(1+r)^t} \right) \forall\, t \in\ T \tag{1}$$

The NPV is calculated by summing up the discounted cumulative cash flows ($dCCF_t$) over the planning periods over 30 years (T), from the initial period t_0(construction start) to year 30, with certain discount rate of 10% (r) as shown in Eq. 2.

$$dCCF_t = \sum_{t=t_0}^{T} aftaxCF_t - CAPEX_t \tag{2}$$

The discounted cash flows ($dCCF_t$) at time (t), can be calculated by subtraction of the after-tax cash flow ($aftaxCF_t$) at time (t) and the capital expenditures ($CAPEX_t$) within 3 years period of construction, which represents the overall capital investment required. The $aftaxCF_t$, is given by the Eqs. 3 – 7.

$$GRev_t = Revenue_t - OPEX_t \; \forall \; t \in T \tag{3}$$

$$ProfitBeforeTax_t = GRev_t - D_t \; \forall \; t \in T \tag{4}$$

$$InCTax_t = ProfitBeforeTax_t \times Tax_t \; \forall \; t \in T \tag{5}$$

$$NetRev_t = ProfitBeforeTax_t - InCTax_t \; \forall \; t \in T \tag{6}$$

$$aftaxCF_t = NetRev_t + D_t \; \forall \; t \in T \tag{7}$$

Where, $GRev_t$ is gross revenue of processes, defined as the total revenue $Revenue_t$ minus operational expenditure ($OPEX_t$). The annual expenses, including the $OPEX_t$ and depreciation charge (D_t). The income tax ($InCTax_t$) is determined by the profit before-tax ($ProfitBeforeTax_t$) multiplied by the UK taxation rate (Tax_t). The taxation charge has to be applied only when a positive annual gross profit is obtained, otherwise it must be avoided. The taxation rate (set equal to 25%, which represents a conservative approximation with respect to the current the UK taxation). $NetRev_t$ is the net revenue after tax, comprises of the profit before-tax minus income tax. $aftaxCF_t$ is the cash flow after tax, the sum of net revenue after tax ($NetRev_t$) and depreciation of charge.

The evaluation of CAPEX, resulting from the sum of the total equipment purchased cost ($TEPC$), total plant direct cost ($TPDC$), total plant indirect cost ($TPIC$), fixed capital investment (FCI), working capital (WC), and land use (LU), as given in Eq. 8.

$$CAPEX = \sum_{t=1}^{T} TEPC + TPDC + TPIC + FCI + WC + LU \tag{8}$$

The equipment purchased cost ($EC_{j,k}^{Equipment}$) of unit operation k at process technology j can be calculated using Eq. (9). EC_k^{base} is purchase cost of unit operation k in the base case, $F_{n,j,k}$ is the sum of mass flow rate of component n to process technology j at operation unit k, F_k^{base} is the mass flow rate of unit operation k in the base case, and θ is sizing factor. The $TEPC$ is calculated by summation of all purchased technology process j at operation unit k using Eq. (10).

$$EC_{j,k}^{Equipment} = EC_k^{base} \left(\frac{\sum_n F_{n,j,k}}{F_k^{base}} \right)^{\theta} \tag{9}$$

$$TPEC = \sum_j \sum_k EC_{j,k}^{Equipment} \tag{10}$$

The $OPEX$ is derived through the sum of the direct manufacturing cost ($DirC$), fixed manufacturing cost ($FixC$), and general manufacturing cost ($GenC$) seen by Eq. 11.

$$OPEX = DirC + FixC + GenC \tag{11}$$

The total raw material cost (RMC) is determined as sum of the market cost of BW used (C_{BW}) and market cost of chemicals consumed (C_c^{Chem}) times with flow rate of BW supplied ($F_{BWn,,j}$) and flow rate of chemicals consumed in the processing stages ($F_{c,j}$). The total utilities cost (UTC) is calculated by the summation of consumption of utilities C_u^{Util} times the flow of utilities $F_{u,j,k}^{Util}$ in process technology j at unit operation k. The direct manufacturing cost, is defined as the sum of the raw material cost (RMC), utilities cost (UTC), operating labor (OLC), supervisory and clerical labor ($SupC$), maintenance and repair ($ManC$), operating supplies (OpC), laboratory charges ($LabC$), patents and royalties (PaC), as expressed in Eqs 12 -14.

$$RMC = \sum_n \sum_j C_{BW} \times F_{BWn,j} + \sum_c \sum_j C_c^{Chem} \times F_{c,j} \ \forall c \in C; j \in J \tag{12}$$

$$UTC = \sum_u \sum_j \sum_k C_u^{Util} \times F_{u,j,k}^{Util} \ \forall u \in U; j \in J \tag{13}$$

$$DirC = RMC + UTC + OLC + SupC + ManC + OpC + LabC + PaC \tag{14}$$

Term $FixC$ is determined by Eq. 15, which is the sum of the local taxes and insurance ($TaxinC$) and plant overhead ($PlantC$).

$$FixC = TaxinC + PlantC \tag{15}$$

$GenC$ is calculated by Eq. 16, as it represents the sum of administration cost ($AdmC$), distribution and selling cost ($DisC$), and research and development ($RnDC$).

$$GenC = AdmC + DisC + RnDC \tag{16}$$

The total final products revenue ($Prod_o^{rev}$) generated by Eq. 17 where all market products (o) is given by the product of selling price ($\mu_{o,j}$) of each product and the flow of products ($F_{o,j}^{Prod}$) from the process technology j.

$$Prod_o^{rev} = \sum_o \sum_j \mu_{o,j} \times F_{o,j}^{Prod} \tag{17}$$

The total co-products revenue ($Coprod_q^{rev}$) is given by Eq. 18, as the co-products (q) of the selling price ($\mu_{q,j}$) , and the co-products market flows ($F_{q,j}^{Coprod}$) from the process technology j.

$$Coprod_q^{rev} = \sum_q \sum_p \mu_{q,j} \times F_{q,j}^{Coprod} \tag{18}$$

The total revenue is equal to the revenue of product produced ($Prod_o^{rev}$) and co-products produced ($Coprod_q^{rev}$), this is ensured by Eq. 19

$$Revenue = \sum_o Prod_o^{rev} + \sum_q Coprod_q^{rev} \tag{19}$$

2.2.2. Environmental objective

This second objective aims to maximize GWP savings related to the IBWR system by adhering to the principles and standards set forth in the ISO 14040/14044 LCA framework. In this work, we integrate LCA techniques with a mathematical programming approach to develop a systematic method for the automatic generation and assessment of processes. The life cycle stages considered include grinding, pretreatment, glucose upgrading, lipid upgrading, protein extraction, AD, incineration, and CCU. Emission data for each process within the selected life cycle stages are collected and sourced from the Ecoinvent database 3.8, as shown in the Table A9 and A10. For the

environmental dimension, GWP is chosen as the primary indicator because they represent a globally recognized and policy-relevant metric for assessing environmental sustainability (Akyuz & Ekmekci, 2025; Hafyan et al., 2020). The environmental objective function $GWPsavings$ in Eq. (20) can be expressed as the sum of the emission credit ($GWP^{Credits}$) obtained from the products and coproducts minus the emissions generated at all stages of the IBWR system ($GWP^{Emissions}$).

$$Max\ Z_2(GWPsavings) = GWP^{Credits} - GWP^{Emissions} \tag{20}$$

$$GWP^{Credits} = \sum_{o}\sum_{q}\sum_{j}(\epsilon_{o,j}^{Prod} \times F_{o,j}^{Prod} + \epsilon_{q,j}^{CoProd} \times F_{q,j}^{Coprod}) \tag{21}$$

$$GWP^{Emissions} = \sum_{c}\sum_{j}\epsilon_{c,j}^{Materials} \times F_{c,j} + \sum_{u}\sum_{j}\epsilon_{u,j}^{Utilities} \times F_{u,j} \tag{22}$$

In Eq. (21), the model for emissions associated with environmental credits ($GWP^{Credits}$) is derived from the credits due to emissions avoided. These avoided emissions factor ($\epsilon_{o,j}^{Prod}, \epsilon_{q,j}^{CoProd}$) result from the displacement of both products and co-products ($F_{o,j}^{Prod}, F_{q,j}^{Coprod}$) generated in the system. The avoided products used for credit allocation are fossil-based products. UK and European data are used for product and co-product inputs where available, while global data are applied when specific datasets are lacking. The avoided products considered in this study are clearly listed in Table A10. In contrast, Eq. (22) calculates the induced emissions ($GWP^{Emissions}$) as the sum of emissions from two sources: (1) emissions from the consumption of chemicals (NaOH, methanol, yeast, nutrients, etc), and (2) emissions from utilities consumption using in both intermediate processing stages and final product processing stages. Each of these emissions is determined by multiplying the emission factor ($\epsilon_{c,j}^{Materials}, \epsilon_{c,j}^{Utilities}$) by their respective associated flow rates ($F_{c,j}, F_{u,j}$). Notably, the foreground data underlying these calculations, which includes process-specific flows and consumptions, are presented in detail in Table A11.

2.2.3. Social impact objective

The social impact (SI) objective of this model is to maximize social benefit by quantifying the employment creation potential within the IBWR system. In this study, employability is used as a representative social indicator to demonstrate how social aspects can be incorporated into the optimization framework. Employment generation is selected because it is a measurable, data-driven parameter that reflects the socioeconomic benefits of industrial biorefinery development, particularly in the context of circular economy initiatives (Manhongo et al., 2025). Other social indicators, such as community well-being, worker safety, and social acceptance, are not included here, but they are discussed in detail elsewhere (Sadhukhan et al., 2021). Employment creation is categorized into three types: direct, indirect, and induced. Direct jobs are those directly employed by the project, while indirect jobs are those created in the supply chain providing inputs for the project. Induced jobs arise from increased consumption driven by both direct and indirect workers (Chazara et al., 2017). A common indicator used in assessing green stimulus projects or programs is the number of jobs created per US dollar (USD) invested (Hanna et al., 2024). The employment multiplier factors covering induced, indirect, and direct job creation are adopted from the report by Bacon and Kojima (2011), which analyzes job creation associated with spending in the alternative energy sector under stimulus-type programs. The mathematical model for employment creation integrates investment, job generation multipliers, and revenue contributions to estimate total employment opportunities across biomass-based processes, as seen in Eq. (23). For each stage j, the total capital investment (TCI_j) is multiplied by the multiplier job factor of biomass-based feedstock ($\bar{Q}_{biomass}$) and the dimensionless revenue ratio

for process ($\vartheta_j^{revenue}$). The dimensionless ratio between individual product revenues and total revenues as the key differentiator between chemical product options (Eq. 24).

$$Max\ Z_3(SI) = \sum_k TCI_{j,k}^{Incineration} \times \bar{Q}_{biomass} \times \vartheta_{Incineration}^{revenue} + \sum_k TCI_{j,k}^{AD} \times \bar{Q}_{biomass} \times \vartheta_{AD}^{revenue} + \sum_k TCI_{j,k}^{Biorefinery} \times \bar{Q}_{biomass} \times \vartheta_{Biorefinery}^{revenue} + \sum_k TCI_{j,k}^{CCU} \times \bar{Q}_{biomass} \times \vartheta_{CCU}^{revenue} \quad (23)$$

$$\vartheta_j^{revenue} = \frac{Revenue\ of\ process\ technology, j}{Total\ revenue\ from\ all\ processes\ of\ IBWR} \quad (24)$$

2.2.4. Constraints

2.2.4.1. BW supply and availability

$$\sum_n FBW_{n,j} \leq SBW_n\ \forall\ n \in N \quad (25)$$

Eq. (25) assures that, the amount of BW ($FBW_{n,j}$) for process stages j is restricted by the amount of BW available (SBW_n).

2.2.4.2. Mass balance constraints

The superstructure is built by definition and interaction of the different unit-operations. The generic mass balance interaction of the unit operations is depicted in Figure S1. Mass balance at each processing stage must be satisfied. Linear modelling is performed for mass balance constraints. The balance equation for mixers is as follows Eq.26.

$$F_{n,j,k}^{IN} = FW_{n,j,k} + \sum_c F_{c,j,k}^{C} + \sum_{n'} F_{n,j,k}^{R} \quad (26)$$

Where, the outlet flow rate of each component n in the mixer, denoted as $F_{n,j,k}^{IN}$, is equal to the sum of the flow rates from the various inlet streams. These inlet streams include: (1) the feed stream, represented by $FW_{n,j,k}$, which carries component n into the mixer at stage j and unit operation k; (2) the additional input streams such as chemicals or additives, where the flow of each chemical component c into the mixer is represented by $F_{c,j,k}^{C}$; and (3) the recovery stream, represented by $F_{n,j,k}^{R}$ which carries component n from a different stage j and unit k. Here, $F_{(n,j,k)}^{R}$represents materials recovered from other stages or units. Specifically, it can include: 1) organic residuals from biorefinery processes that are mandatorily digested in AD, and 2) Solid residues that are recovered for incineration. The reaction can take place in the reactor unit, which can be calculated based on the conversion rate and stoichiometric coefficient of the inlet flows consumed (Eq.27).

$$F_{n,j,k}^{OUT} = F_{n,j,k}^{IN} + \sum_r \sum_m F_{m,j,k}^{IN} \times\ \alpha_{n,r,j,k} \times X_{m,r,j,k} \quad (27)$$

Where $F_{m,j,k}^{OUT}$ is the outlet flow rate of component n from the reactor at stage j and unit operation k, is equal to the inlet flow rate ($F_{n,j,k}^{IN}$), which represents the amount for component n entering the reactor plus the inlet flow rate ($F_{m,j,k}^{IN}$) of reactant m into the reactor. The reaction is calculated by summing the contribution of each reaction r, where each reactant m contributes according to its inlet flow ($F_{m,j,k}^{IN}$), the stoichiometric coefficient $(\alpha_{n,r,j,k})$ of component n in reaction r, and the conversion rate ($X_{m,r,j,k}$) , which represents the fraction of reactant m that is consumed in the reaction. The stoichiometric coefficient $(\alpha_{n,r,j,k})$ can be negative if component n is a reactant

(indicating consumption) or positive if n is a product (indicating formation). Meanwhile, the mass balance correlation for unit operation such as pump, heater, cooler, and compressor where the total mass flow into any process unit must equal the total mass flow out can be written as Eq.28.

$$\sum_{n} F_{n,j,k}^{OUT} = \sum_{n} F_{n,j,k}^{IN} \ \forall\, n, j, k \tag{28}$$

Eq.(28) ensures that the flow in equals the flow out for each unit operation k at each process stage j for each component n. The mass balance for a separation unit can be described by Eq. (29) that ensures the total amount of each component entering the unit is equal to the sum of all the outlet streams.

$$F_{n,j,k}^{OUT} = F_{o,j,k}^{Prod} + F_{q,j,k}^{CoProd} + F_{n,j,k}^{W} + F_{n,j,k}^{R} \tag{29}$$

The inlet flow of each component n into the separation unit is divided into four potential outlet streams: the product stream, represented as $F_{o,j,k}^{Prod}$, which carries the desired product; the co-product stream, denoted $F_{q,j,k}^{CoProd}$, which represents a secondary valuable output; the waste stream, represented $F_{n,j,k}^{W}$, which contains unusable material; and the recovery stream, denoted $F_{n,j,k}^{R}$, which returns material to a previous stage or unit for further processing.

2.2.4.3. Utilities consumption

This model considers three utilities: electricity, steam, and cooling water. For unit operation k in technology j, the energy demand for electricity, heating, and cooling is determined using energy balances similar to those applied in mass balances. The electricity consumed by unit k in technology j is calculated by multiplying the relevant mass flow ($F_{n,j,k}^{EL}$) by the specific electricity consumption ($\delta_{j,k}^{EL}$), as seen in Eq. (30).

$$EL_{j,k}^{Consumed} = F_{n,j,k}^{EL} \cdot \delta_{j,k}^{EL} \tag{30}$$

The electricity production results from the combustion of BW and solid waste (SW). This generated electricity is intended to decrease electricity demand, thereby reducing operating costs and minimizing environmental impact. The amount of electricity produced ($EL_{j,k}^{Prod}$) is calculated based on the mass flow of the combusted BW (F_{BW}^{IN}) and SW (F_{SW}^{IN}), as well as their lower heating value (LHV) (LHV_{BW}, LHV_{SW}) and the efficiency of the turbine (γ^{TUR}). Electricity production is expected to meet all electricity consumption needs, resulting in a surplus of electricity ($EL_{j,k}^{EXCESS}$) that can be sold to the electricity grid (Eq.31).

$$EL_{j,k}^{Prod} = (F_{BW}^{IN} \cdot LHV_{BW} + F_{SW}^{IN} \cdot LHV_{SW}) \cdot \gamma^{TUR} \tag{31}$$

$$EL_{j,k}^{EXCESS} = \sum_{j,k} (EL_{j,k}^{Prod} - EL_{j,k}^{Consumed}) \tag{32}$$

Both steam and cooling consumed ($H_{j,k}^{Consumed}, C_{j,k}^{Consumed}$) are calculated in a similar manner to the electricity demand, using the relevant mass flow ($F_{n,j,k}^{H}, F_{n,j,k}^{C}$) multiplied by their respective specific heat and cooling water consumption ($\delta_{j,k}^{H}, \delta_{j,k}^{C}$) (Eqs. 33 – 34).

$$H_{j,k}^{Consumed} = F_{n,j,k}^{H} \cdot \delta_{j,k}^{H} \tag{33}$$

$$C_{j,k}^{Consumed} = F_{n,j,k}^{C} \cdot \delta_{j,k}^{C} \tag{34}$$

The produced steam ($H_{j,k}^{Prod}$) is calculated based on the mass flow of combusted BW and SW, as well as their respective LHV and turbine efficiency, divided by the specific heat enthalpy of high-

pressure steam (Cp_{heat}). Additionally, a specific condition is established, requiring that steam production must meet the total steam demand for all selected optimal technologies (Eq. 35).

$$H_{j,k}^{Prod} = \frac{(F_{BW}^{IN} . LHV_{BW} + F_{SW}^{IN} . LHV_{SW}) . \mathfrak{r}^{TUR}}{Cp_{heat}} \tag{35}$$

$$\sum_{i} H_{j,k}^{Prod} - \sum_{i} H_{j,k}^{Consumed} = 0 \tag{36}$$

2.2.4.4. Decision variables

The optimization involves two types of decision variables. Continuous variables represent the proportion of $FBW_{n,j,k}$ allocated to each stage (enzymatic hydrolysis, AD, and incineration), while binary variables determine the selection processes for glucose upgrading (bioethanol, 2,3-BDO, butanol, LA and SA). The flowrate of BW, for component n at stage j is determined by the proportions of $FBW_{n,j,k}$ allocated to each stage, represented by the continuous variables (x_j), corresponding to enzymatic hydrolysis, AD, and incineration, respectively (Eq. 37).

$$FBW_{n,j}^{TOTAL} = FBW_{n,j} \times x_j \tag{37}$$

$$x_1 + x_2 + x_3 = 1 \ \forall x_1, x_2, x_3 \in 0 \leq x_j \leq 1 \tag{38}$$

The continuous variables (x_1, x_2, x_3) indicate the portion of FWB distributed to three different technologies. A key constraint is that the sum of these proportions must be equal to 1, as expressed in Eq. (38). On the other hand, the binary variable is defined using big M formulation (Eq. 39). This approach ensures that the flowrates of glucose ($F_{n,j,k}$) are activated only when their corresponding binary decision variables (y_n) are set to 1, thus maintaining control over the selection process for glucose products. The binary decision variables are applied to the Glucose Upgrading. After enzymatic hydrolysis, glucose can be upgraded into one of five products (bioethanol, 2,3-BDO, butanol, LA or SA). The binary variable (y_n) (Eq. 40), ensures that only one product is chosen for production. $y_j = 1$: if glucose is used for product (o); otherwise $y_j = 0$.

$$0 \leq F_{n,j,k} \leq M \cdot y_j \tag{39}$$

$$y_1 + y_2 + y_3 + y_4 + y_5 \leq 1 \ \forall y_1, y_2, y_3, y_4, y_5 \in \{0,1\} \tag{40}$$

2.2.4.5. Demand constraints

The amount of products produced is bounded by the demands, as shown in Eq. 41.

$$F_{o,j,k}^{Prod} \times y_j \leq D_o^{max} \ \forall \ o \in O \tag{41}$$

This ensures that if a product is selected (i.e., $y_j = 1$), the flowrate for product produced ($F_{o,j,k}^{Prod}$) lies within the specified maximum demands (D_o^{max}). This is to ensure the production does not exceed the demand. Details of the demand for glucose-based products are provided in Table A5.

2.2.4.6. Processing limits

The processing limits are set to ensure that the IBWR system must be at least 100 MT/day of BW is consumed as shown in Eq.42.

$$\sum_{n} FBW_{n,j} \geq BW_n^{min} \ \forall \ n \in N \tag{42}$$

2.2.5. *Multi-objective optimization problem formulation*

To establish an optimal waste management policy, a multi-objective approach is adopted that considers economic, environmental, and social factors. Because these objectives often conflict, decision-making becomes complex. The ε-constraint method is employed to address this challenge by converting the multi-objective problem into a single-objective optimization problem. In this method, one objective function (e.g., NPV) is maximized, while the other objectives are expressed as constraints limited by user-defined ε values. The allowable bounds (ε-min and ε-max) for each secondary objective, such as GWP saving and employability, are first determined by independently minimizing and maximizing each of them under the same model constraints. These bounds are then used to explore trade-offs and generate the Pareto-optimal frontier. This approach provides clearer insight into the interdependencies among objectives and offers advantages over traditional weighting techniques(Cambero & Sowlati, 2016). In summary, the resulting formulation for the MOO problem of the IBWR system can be mathematically expressed by Eq. 43.

$$Max\ NPV \tag{43}$$

Subject to:

$\epsilon_{GWPmin} \leq GWP\ savings\ \leq\ \epsilon_{GWPmax}$

$\epsilon_{SImin} \leq SI\ \leq\ \epsilon_{SImax}$

Capital investment

Operating cost

Revenue

Mass balance

Utility consumption

$BW_n^{min} \leq FBW_{n,j,k} \leq SBW_n$

$F_{o,j,k}^{Prod} \times y_j \leq D_o^{max}$

Non-Negativity constraints

$x_j = \{0 - 1\}$

$y_j = \{0,1\}$

$FBW_{n,j,k} \geq 0$

$F_{n,j,k} \geq 0$

$F_{c,j,k}^{C} \geq 0$

$F_{o,j,k}^{Prod} \geq 0$

$F_{q,j,k}^{CoProd} \geq 0$

Where ϵ_{GWP} and ϵ_{SI} are the allowable limits for the environmental and social objectives, respectively.

2.2.6. *Model solution*

The superstructural model is implemented in Python version 3.12. This model represents a large-scale MINLP problem, incorporating constraints for mass and energy balances, capital and operating costs, feedstock availability, product demand, and specific objective functions. The constraints are non-convex, which can hinder solution convergence and make finding a global optimum difficult. The model was solved using the pymoo platform, which employs the Solving Constraint Integer Programs (SCIP) solver. SCIP, a branch-and-bound-based solver, is particularly effective for nonlinear

optimization problems (Uebbing et al., 2020). All model runs were conducted on a system with a 1.30 GHz 12th Gen Intel(R) Core(TM) i5-1235U processor, 16 GB of RAM, and a 64-bit operating system.

2.2.7. Case studies

This work focuses on exploring three distinct heat supply scenarios for IBWR system, aiming to determine how different strategies for heat consumption influence the overall system configuration and performance. These case studies demonstrate the applicability of the proposed mathematical programming model in identifying the optimal pathway configuration for a biorefinery producing multiple products. Heat supply scenarios must be carefully analyzed because the preheat, separation, and purification processes of biochemical and biofuel production are highly heat-intensive. Meeting these energy demands significantly impacts the BW allocation within the plant. Specifically, fulfilling heat requirements often necessitates diverting a portion of the BW for incineration, which in turn affects the overall configuration and resource distribution in the IBWR system. Despite this, the system ensures that electricity production remains sufficient to meet internal consumption needs, with any excess electricity generated being sold to the grid as an additional revenue stream. This consideration of heat supply underscores the importance of evaluating different strategies for balancing energy generation and consumption. The three heat supply scenarios evaluated in this study are as follows:

- **Case 1 (Internal heat generation only):** In this scenario, all the heat required by the IBWR system is produced internally using available resources. No external heat is utilized.
- **Case 2 (Partial heat supply):** In this scenario, the plant relies on a combination of internally generated heat and supplementary heat sourced externally. The external supply meets any shortfall in heat requirements that cannot be fulfilled internally.
- **Case 3 (Full heat supply):** In this scenario, 100% of the heat required by the IBWR system is sourced externally. The plant does not generate any heat internally; instead, it purchases the entire heat demand from external suppliers.

Analyzing these scenarios helps in evaluating the implications of different energy supply strategies on the optimal design and operational configuration of the IBWR system. This comparison provides insights into the trade-offs between internal energy generation, external energy procurement, and their impacts on system performance and cost efficiency.

3. Results

The analyses conducted based on different case studies are explored through two distinct approaches. The first approach focuses on a single-objective framework, where each objective is evaluated individually to observe its standalone results. This initial step provides insights into the outcomes when only one objective is prioritized. In the second approach, a multi-objective analysis is performed to examine the trade-offs between multiple objectives, specifically economic, environmental, and social impacts.

3.1. Single-objective optimization results

The single-objective analysis is conducted through multiple runs across different case studies, with each run focusing on optimizing a specific objective independently. In the first run, the solution is obtained by exclusively maximizing the NPV to assess the economic performance. The second run focuses solely on maximizing the GWP savings, emphasizing the environmental benefits. Finally, the third run is dedicated to maximizing employability generation, addressing the social impact.

As presented in Table 1, the analysis of various cases highlights distinct optimal configurations for the IBWR system, with these differences driven by the energy strategies employed. When the primary goal is to maximize NPV, Case 3 achieves the highest NPV of 733 M$, significantly outperforming Case

1 (291 M$) and Case 2 (555 M$). Figure 3a illustrates this superior performance, highlighting that the elevated NPV in Case 3 arises mainly from its high annual revenue of 553 M$/year. Although the configuration in Case 3 involves a high investment (263 M$) and operational cost (383 M$/year), the system achieves a substantial economic margin, which ultimately leads to its higher NPV. A key contributing factor is the reliance on fully external heat sources, which allows for the full allocation of BW to biorefinery pathways. This strategy focuses on producing more marketable products instead of using BW for incineration and AD, thereby enhancing profitability. Across all evaluated, LA emerges as the primary among glucose-based products. LA is preferred because of its superior market price, high demand, and competitive yield of 0.46 kg per kg of BW (Mailaram et al., 2023). Importantly, sensitivity analysis indicates that LA remains the dominant product even under ±20% price variations, continuing to contribute the largest share to the NPV. This robustness to price fluctuations is illustrated in Figure S2.

Additionally, the imposed constraint of 800 MT/day of optimal BW supply enables the production of approximately 291.17 MT/day of LA, generating significant revenue. Other sources of revenue come from additional products and co-products, as detailed in Table 1. Although BW is not primarily allocated to incineration and AD, these processes still play a role in managing residuals; organic remnants from LA other selected processes are processed through AD to produce biomethane (6.36 MT/day), while solid waste from protein extraction is incinerated to generate excess electricity (2,196 kWh/day). In contrast to Case 3, Cases 1 and 2 exhibit a comparable optimal configuration of the IBWR system. In both cases, where heat demand is either fully or partially required, BW is primarily allocated to incineration to meet thermal energy needs, with the remaining portion directed toward the biorefinery pathway to maximize economic returns. In Case 1, approximately 24% of the available BW is directed to incineration, fully satisfying the internal heat demand of 310,175 MJ/hr and enabling complete energy self-sufficiency. The remaining 76% is utilized in the biorefinery route to produce LA (220.57 MT/day) as the main product, alongside biodiesel (11.4 MT/day), protein (55.38 MT/day), glycerol (0.85 MT/day), biomethane (4.82 (MT/day), fertilizer (6.94 MT/day) and CO_2-captured (249.21 MT/day). The system also generates an electricity surplus of 10,061 kWh/day, enhancing its energy autonomy. Economically, Case 1 yields a NPV of 291 M$, with a capital investment of 269 M$, operating cost of 354 M$/year, and revenue of 441 M$/year. Although energy self-sufficiency is achieved, the higher share of BW used for incineration limits the potential for high-value bioproduct generation, constraining economic performance. In Case 2, the fraction of BW directed to incineration decreases to 10%, while 90% is processed through the biorefinery pathway. This reduction in incineration lowers internal heat generation (178,633 MJ/hr), making the system partially dependent on external heat sources. Nevertheless, the increased biorefinery throughput significantly enhances revenue generation, reaching 507 M$/year, which offsets the slightly higher operating cost of 370 M$/year. The required capital investment also decreases marginally to 268 M$ due to smaller incineration infrastructure. A similar IBWR system configuration for Case 2 is summarized in Table 1.

Focusing the optimization on maximizing GWP savings highlights the distinct environmental performances of the IBWR system configurations. The analysis of the model aimed at maximizing GWP savings demonstrates that Case 1 represents the most environmentally efficient solution, achieving the highest GWP savings of 1,086 tCO_2-eq /day. This exceptional performance is primarily due to the fact that all the required heat energy in Case 1 is supplied through internal heat generation. In comparison, Case 2 and Case 3 yield considerably lower GWP savings, at 817 tCO_2-eq/day and 422 tCO_2-eq/day, respectively. A key factor contributing to the substantial GWP reduction in Case 1 is the effective heat consumption reduction strategy, which also plays a notable role in Case 2. The configuration of the IBWR system in Case 1 further distinguishes it from the other cases. Specifically, 24% of the BW is directed to incineration, while the remaining 76% is allocated to the biorefinery route. When partial heat is supplied from external sources, the share of BW allocated to incineration

decreases further to 10%, with a corresponding increase in allocation to the biorefinery route up to 90%. In contrast, Case 3 allocates 100% of BW to the biorefinery pathway.

A notable difference in GWP performance among the cases is evident from their heat consumption profiles, as shown in Figure 3b. In all cases, the GWP credit, representing avoided fossil-based emissions, is the dominant contributor to total savings. In Case 1, the GWP credit amounts to 1,123 tCO_2-eq/day, comprising approximately 97% of the total GWP savings, with minor contributions from chemical use (2%) and cooling water (1%). In Case 2, the GWP credit share increases slightly; however, total GWP savings decrease to about 74% of those in Case 1 due to partial reliance on externally supplied heat. Heat-related emissions represent around 23% of the total, while chemical use and cooling water contribute approximately 2% and 1%, respectively. In Case 3, although the GWP credit remains the highest (around 60%), overall GWP savings decline substantially as heat-related emissions rise to about 38% of the total, with chemical use and cooling water contributing a further 1% combined, indicating the greater environmental burden associated with full external heat supply. Across all scenarios, the IBWR system maintains a similar optimal configuration as observed in the NPV-focused analysis and is designed to utilize a maximum BW supply of 800 MT/day to achieve optimal operational efficiency. The biorefinery pathway consistently serves as the primary route for BW valorization, with LA produced as the main product due to its high yield and strong GWP credit. In addition, co-products such as biodiesel, protein, glycerol, biomethane, fertilizer, and captured CO_2 are generated consistently across cases, contributing to both the system's economic viability and environmental performance. Furthermore, heat generation and consumption vary with energy strategy: in Case 1, internal heat generation matches consumption (310,175 MJ/hr), achieving energy self-sufficiency; in Case 2, internal heat is lower (178,634 MJ/hr) and partially supplemented by external sources (357,269 MJ/hr total consumption); in Case 3, internal heat is minimal (85,600 MJ/hr) and the system relies fully on external heat (390,576 MJ/hr total). Surplus electricity similarly decreases with increasing external heat dependence: 10,055 kWh/day in Case 1, 5,446 kWh/day in Case 2, and 2,187 kWh/day in Case 3.

When the objective shifts to maximizing employability, Case 3 emerges as the most effective configuration, generating 4,881 jobs compared to 3,337 in Case 1 and 4,597 in Case 2. This outcome is primarily driven by the higher total revenue generated from the biorefinery subsystem, which dominates Case 3's overall income and thus increases its employment potential. This difference reflects the system's configuration and the allocation of BW. In Case 3, the entire BW supply of approximately 800 MT/day is directed to the biorefinery pathway, with no allocation to incineration or AD. In contrast, Cases 1 and 2 distribute BW across pathways: Case 1 allocates roughly 24% to incineration and 76% to the biorefinery, while Case 2 allocates 10% to incineration and 90% to the biorefinery. As illustrated in Figure 3C, the biorefinery pathway is the dominant driver of employment, accounting for over 98% of total jobs in all cases. Contributions from incineration, AD, and CCU are minimal. Consistent with the other objectives, LA remains the primary product across all cases, with co-products including biodiesel, protein, glycerol, biomethane, fertilizer, and captured CO_2 consistently generated. The allocation of BW also affects energy generation and consumption. Case 1 produces 310,175 MJ/hr of heat internally, fully meeting the system's heat demand and resulting in an electricity surplus of 10,010 kWh/day. Case 2 generates 178,634 MJ/hr of heat internally, requires partial external heat supply, and produces electricity with a surplus of 5,327 kWh/day. Case 3 relies almost entirely on external heat (85,600 MJ/hr generated internally), producing the least electricity with a small surplus of 2,067 kWh/day.

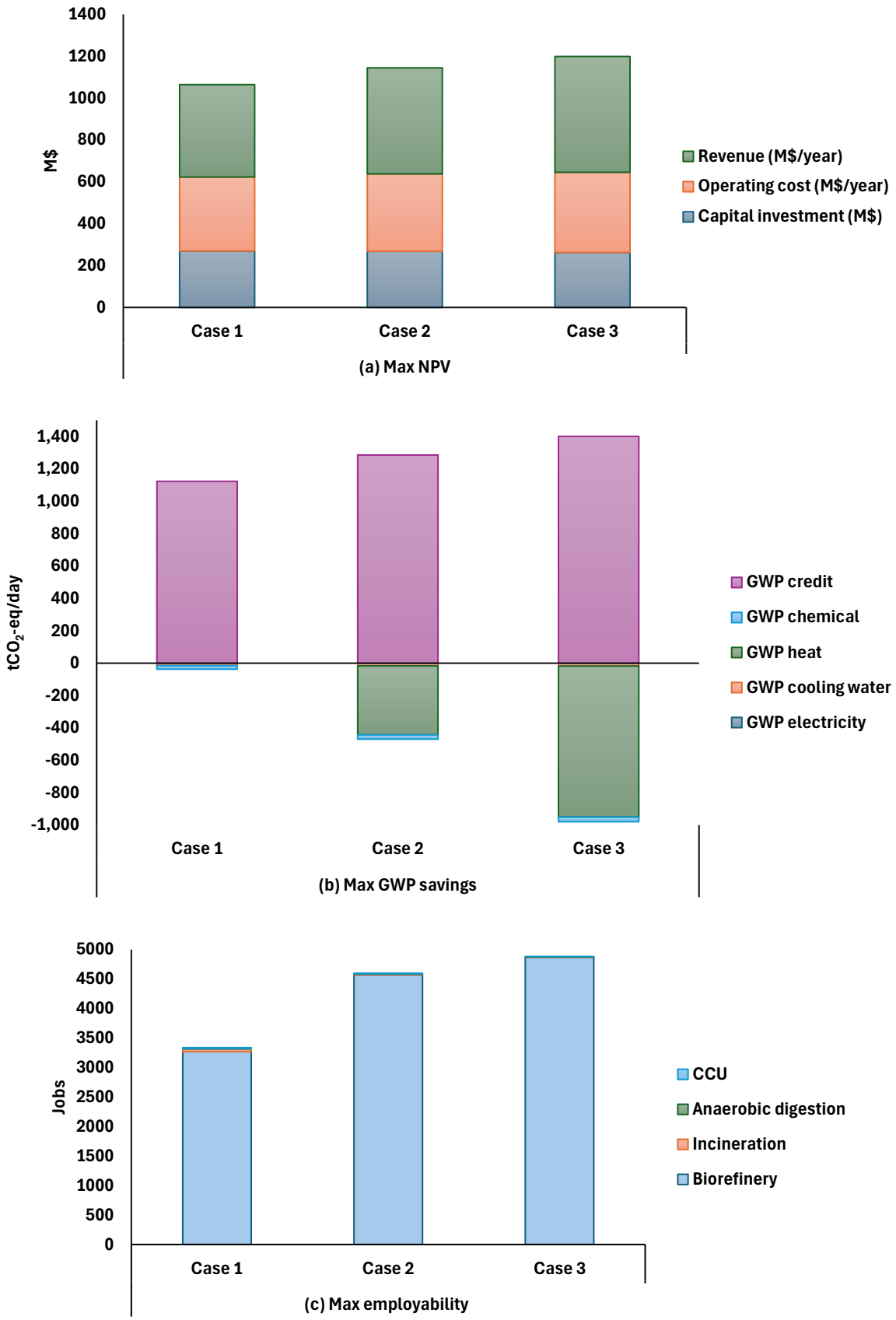

Fig. 3. Percentage contribution of parameters to each objective function: (a) Max NPV; (b) Max GWP savings; and (c) Max employability

Table 1. Optimal solutions of single objectives for all cases studied

Parameter	Max NPV			Max GWP savings			Max Employability			Unit
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	
BW supply	800	800	800	800	800	800	800	800	800	MT/day
NPV	291	555	733	-	-	-	-	-	-	M$
GWP savings	-	-	-	1,086	817	422	-	-	-	tCO_2-eq/day
Employability	-	-	-	-	-	-	3,337	4,597	4,881	Jobs
Stage selected (x_j)										
Incineration	24%	10%	0%	24%	10%	0%	24%	10%	0%	% allocation
AD	0%	0%	0%	0%	0%	0%	0%	0%	0%	% allocation
Biorefinery	76%	90%	100%	76%	90%	100%	76%	90%	100%	% allocation
Glucose selected (y_j)										
Bioethanol	0%	0%	0%	0%	0%	0%	0%	0%	0%	% allocation
BDO	0%	0%	0%	0%	0%	0%	0%	0%	0%	% allocation
Succinic acid	0%	0%	0%	0%	0%	0%	0%	0%	0%	% allocation
Butanol	0%	0%	0%	0%	0%	0%	0%	0%	0%	% allocation
Lactic acid	100%	100%	100%	100%	100%	100%	100%	100%	100%	% allocation
Product & co-products										
Lactic acid	220.57	261.92	291.17	220.57	261.93	291.17	220.57	261.93	291.17	MT/day
Biodiesel	11.40	13.53	15.04	11.40	13.53	15.04	11.40	13.53	15.04	MT/day
Protein	55.38	65.76	73.10	55.38	65.76	73.10	55.38	65.76	73.10	MT/day
Glycerol	0.85	1.01	1.12	0.85	1.01	1.12	0.85	1.01	1.12	MT/day
Biomethane	4.82	5.72	6.36	4.82	5.72	6.36	4.82	5.72	6.36	MT/day
Fertilizer	6.94	8.25	9.17	6.94	8.25	9.17	6.94	8.25	9.17	MT/day
CO_2	249.21	153.86	86.43	249.21	153.86	86.43	249.20	153.86	86.43	MT/day
Heat generated	310,175	178,633	85,600	310,175	178,634	85,600	310,175	178,634	85,600	MJ/hr
Heat consumption	310,175	357,265	134,739	310,175	357,269	390,576	310,175	357,268	390,574	MJ/hr
Electricity generated	11,890	6,848	3,281	11,890	6,848	3,281	11,890	6,848	3,281	kWh/day
Electricity consumption	1,829	1,393	1,085	1,836	1,401	1,094	1,880	1,520	1,215	kWh/day
Electricity excess	10,061	5,454	2,196	10,055	5,446	2,187	10,010	5,327	2,067	kWh/day

3.2. Pareto-optimal solutions for multiple objectives

This subsection presents the results of the MOO analysis, highlighting the trade-offs and synergies among economic, environmental, and social objectives. The aim is to explore how different solutions balance these competing criteria and to identify configurations that achieve sustainable performance across all three dimensions. Within the MOO framework, a set of Pareto-optimal solutions is generated by varying the primary objective function. The performance of each solution is evaluated using three key indicators: economic (NPV), environmental (GWP savings), and social (employability). To support visualization and facilitate the analysis of sustainability trade-offs, Figure 4 presents three distinct perspectives of a three-dimensional scatter plot for each case. In these plots, the x, y, and z axes correspond to the three indicators, with different markers illustrating variations in the third objective. Each point on the Pareto front represents an equally optimal solution, offering a range of trade-off alternatives. Across all case studies, the results reveal a positive correlation among the objectives, as higher NPV values are generally associated with greater GWP savings and increased employability. As one moves upward along the Pareto curve, the simultaneous enhancement of economic, environmental, and social outcomes becomes apparent. Since the optimization framework is designed to maximize all three objectives (NPV, GWP savings, and employability), it effectively identifies solutions that deliver concurrent improvements across these sustainability metrics.

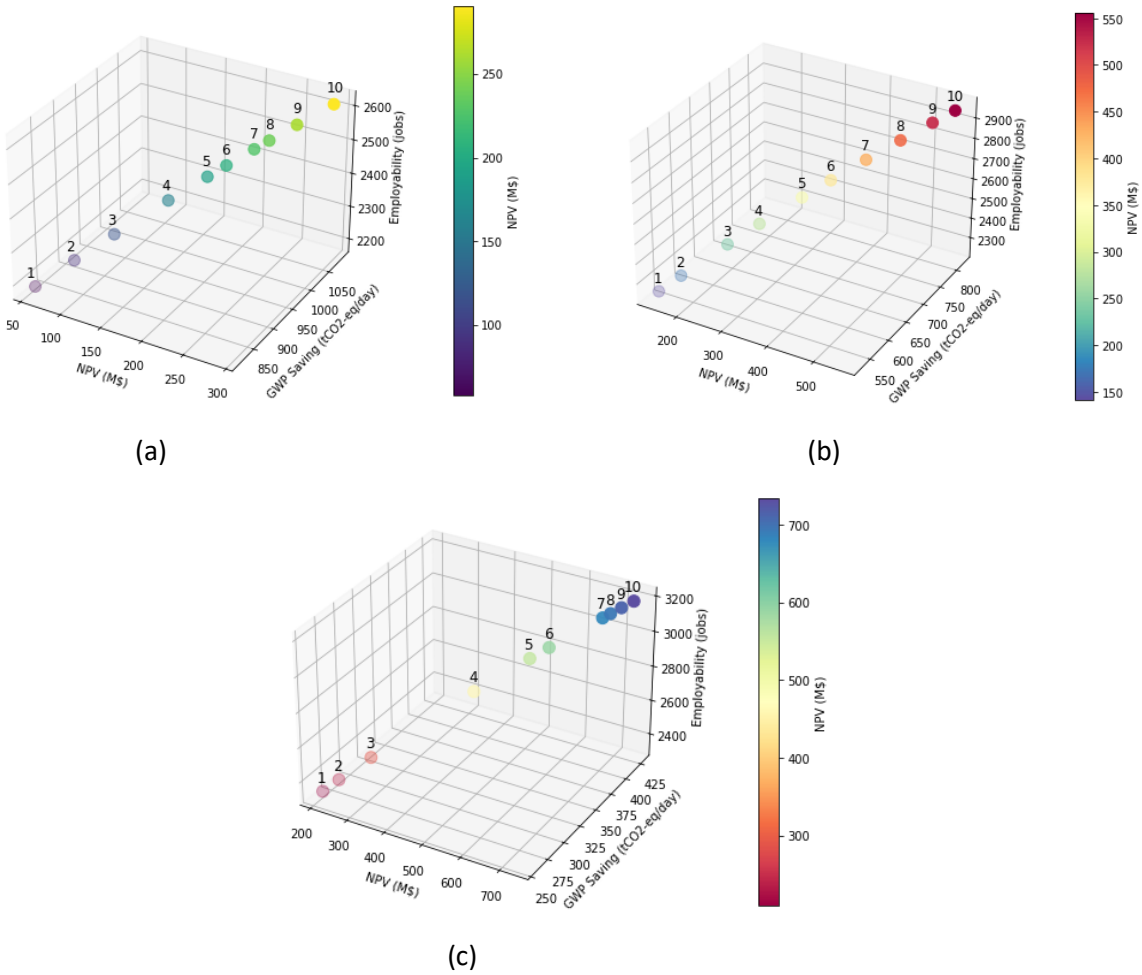

Fig. 4. Pareto fronts for different cases: (a) Case 1, (b) Case 2, and (c) Case 3

A key observation from the analysis is that the optimal BW capacities vary across the Pareto points. As the solution moves from the least economically favorable point to the most optimal one,

BW capacities increase, resulting in improved environmental and social outcomes. The differences in optimal values and system configurations are evident across the cases. To better understand the trade-offs in BW allocation, five representative optimal solutions are selected, as shown in Figure 5. In Case 1, the BW allocation remains consistent across the Pareto front: 24% to incineration and 76% to the biorefinery. The BW supply ranges from 602 MT/day at the least economically optimal point to 800 MT/day at the most optimal point. At the least optimal point, the total NPV is 58 M$, with GWP savings of 816.82 tCO_2-eq/day and 2,190 jobs generated. At the most optimal point, the NPV increases to 291 M$, and GWP savings and employability rise to 1,086.31 tCO_2-eq/day and 2,612 jobs, respectively.

In Case 2, the total BW supply is comparatively lower, ranging from 517 MT/day at the least optimal point to 800 MT/day at the most optimal point. In this case, 90% of the BW is allocated to the biorefinery, with the remaining 10% directed to incineration across all feasible points. Despite the lower BW supply, both the NPV and GWP savings are higher than in Case 1. However, job creation is slightly higher. The least optimal point yields an NPV of 141 M$, GWP savings of 529.28 tCO_2-eq/day, and 2,250 jobs, increasing to 555 M$, 819.36 tCO_2-eq/day, and 2,949 jobs at the most optimal point.

In contrast, Case 3 presents a distinct configuration, where 100% of the BW is allocated to the biorefinery at all Pareto points. The BW supply in this case ranges from 483 MT/day at the least optimal point to 800 MT/day at the most optimal point. At the least point, the NPV is 210 M$, GWP savings are 258.26 tCO_2-eq/day, and 2,334 jobs are generated. Moving upwards, the NPV, GWP savings, and employability improve to 733 M$, 427.43 tCO_2-eq/day, and 3,187 jobs, respectively. Across all cases, the IBWR system configurations are similar at all Pareto points, with LA selected as the optimal glucose-based product, as shown in Figure S3. These findings underscore the interdependence of economic, environmental, and social objectives in achieving sustainable outcomes. The variation in system configurations across the three cases highlights the importance of adopting tailored strategies to effectively balance these competing objectives.

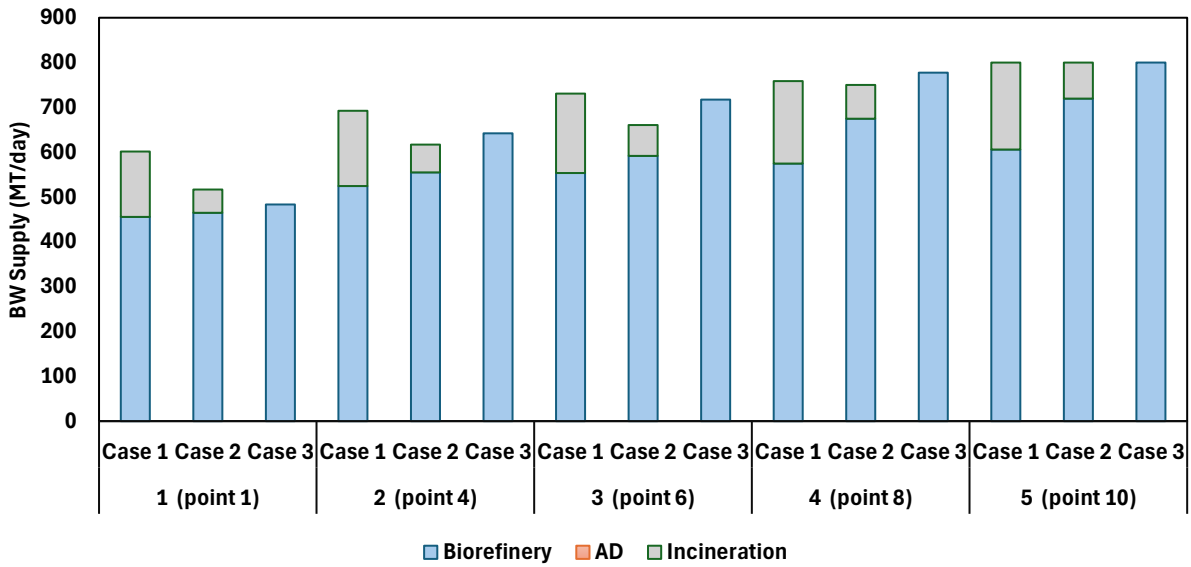

Fig. 5. Optimal BW allocation in the IBWR system across five selected optimal solutions for all cases.

4. Discussions

4.1. Configuration of the proposed IBWR system

Recent results from both single-objective and multi-objective optimization approaches indicate that BW can be effectively diverted to both incineration and biorefinery pathways. This integrated

strategy has the potential to enhance revenue generation while simultaneously reducing FW, thereby supporting the transition toward more sustainable and efficient waste management systems. Currently, in the UK, incineration is the predominant method for FW treatment, accounting for approximately 44% of total waste processed. Other management options include donation (1%), animal feed (23%), recycling via AD and composting (16%), and landfill (16%) (WRAP, 2023). While these figures reflect a reliance on conventional methods, recent studies have explored more strategic sustainable alternatives. Slorach et al. (2020) conducted a comparative economic and environmental assessment of various FW management scenarios in the UK. Their analysis highlighted that a projected 2030 scenario, which allocates 92% of FW to AD, 3% to composting, 4% to incineration, and 1% to landfill, would achieve the lowest environmental impact, at -0.029 tCO_2-eq/t FW, and the lowest treatment cost, at 31.5 $/t FW. Compared to the current system, this optimized scenario could yield an annual reduction of 0.48 tCO_2-eq and cost savings of £338 million. Conversely, a business-as-usual (BAU) scenario, involving 50% incineration, 6% AD, 10% composting, and 34% landfill, was associated with a significantly higher environmental burden (0.073 tCO_2-eq/t FW) and greater cost (126 $/t FW). Another study conducted by Wang et al. (2020) performed an LCA of municipal solid waste (MSW) management in Nottingham, England. Their evaluation of historical and future scenarios revealed that reducing landfill usage and increasing recycling and AD can substantially lower GWP emissions. Scenario 1, dominated by landfill (54.7%) and incineration (40.7%), had the highest GWP of 1.1 tCO_2-eq/t MSW. In contrast, Scenario 3, with only 7.3% landfill, 57.6% incineration, 31.5% composting, and 17.5% recycling, reduced emissions to 0.21 tCO_2-eq/t MSW. The future-oriented Scenario 4, which emphasized AD (31.5%) and further reduced landfill to 7.3%, achieved a net GWP of -0.14 tCO_2-eq/t MSW. The authors suggested to integrate economic and social considerations is essential for a more complete assessment of waste management strategies and policy effectiveness. Although these studies emphasize the need to reduce landfill reliance, they do not offer a holistic assessment encompassing all three pillars of sustainability: environmental, economic, and social. In response to the identified need for an integrated sustainability assessment, this study introduces the IBWR management system, which combines BW valorisation with existing FW management practices. This approach employs a MOO framework that simultaneously considers environmental, economic, and social performance indicators.

Among the Pareto-optimal solutions evaluated, Point 10 yielded the most favorable outcomes across three case studies. The present study introduces the IBWR system that combines BW valorisation through advanced biorefinery and CCU pathways with existing FW treatments (AD and incineration). While the IBWR system exhibits higher treatment costs of 1,360 $/t BW (Case 1), 1,420 $/t BW (Case 2), and 1,470 $/t BW (Case 3), these are largely attributed to the capital and operational demands associated with the incorporation of biorefinery and CCU processes. Nonetheless, these higher costs are accompanied by substantial environmental and economic benefits. The system achieves substantial GWP reductions across all cases, with savings of 1.36 t CO_2-eq/t BW for Case 1, 1.02 t CO_2-eq/t BW for Case 2, and 0.53 t CO_2-eq/t BW for Case 3, notably exceeding the mitigation levels reported in previous studies. Moreover, the IBWR framework enables considerable revenue generation through the production of value-added products, with yields reaching 1,704 $/t BW in Case 1, 1,900 $/t BW in Case 2, and 2,074 $/t BW in Case 3. It is worth noting that the high treatment costs observed in this study reflect a pattern commonly seen in other biorefinery systems. For instance, Galanopoulos et al. (2019) conducted a biorefinery integration scenario for algae valorisation using a photobioreactor with a feedstock capacity of 1.75 kt/year, resulting in a treatment cost of 1,305.43 $/t of algae. Similarly, Wu et al. (2024) evaluated a lignin biorefinery process with a capacity of 2,000 MT/day and reported treatment costs of 578.48 $/t hardwood, 623 $/t softwood, and 484.09 $/t herbaceous biomass. Additionally, Dickson et al. (2021) performed a superstructure optimization study for SA production from biomass at a scale of 30,000 tonnes per year, finding treatment costs as

high as 1,418.3 $/t. These comparisons suggest that the treatment costs reported in the current study align well with those of other advanced biorefinery processes. While the financial demands are notable, they are offset by the significant environmental gains and the potential for high-value product recovery. Therefore, the IBWR system offers a compelling model for sustainable waste management, balancing upfront investment with long-term environmental and economic returns.

When applied across the UK as a whole, the IBWR system could generate substantial climate benefits. Based on the UK's current BW generation of 292,000 tonnes per year, associated with 584,000 tCO_2-eq emissions (Narisetty et al., 2021). The optimized IBWR configurations demonstrate potential annual GWP savings of 361,741 tCO_2-eq/year (Case 1), 272,846 tCO_2-eq/year (Case 2), and 142,334 tCO_2-eq/year (Case 3). This indicates that the IBWR system could offset roughly 68%, 48%, and 26% of the total BW-related emissions , effectively compensating for a large share of the emissions currently generated by BW disposal. Beyond its climate benefits, the IBWR system also transforms the economic outlook of FW management. At present, BW contributes to substantial national losses, which is estimated at £21 billion ($27.63 billion) annually, due to lost product value, environmental degradation, and disposal costs. By contrast, implementing the IBWR framework converts this liability into an economic opportunity, yielding projected annual profits of 78 M$ (Case 1), 128 M$ (Case 2), and 161 M$ (Case 3). This shift represents a transition from an annual economic burden of £21 billion to a system capable of generating up to $125 million in profit, underscoring the IBWR's dual role in mitigating emissions and enhancing economic resilience. Furthermore, the IBWR system is estimated to create 2,612 jobs in Case 1, 2,949 jobs in Case 2, and 3,187 jobs in Case 3. This contribution supports the UK's projected green job growth of 605,000 positions by 2030 (WPI, 2022), where such jobs are defined as those that promote environmental sustainability through emissions reduction, waste minimization, and resource efficiency. Globally, FW estimates that food loss and wase costs the global economy $936 billion/year and generates an estimated 8 to 10% of global greenhouse gas emissions or around 3.3 billion tCO_2-eq/year (Hannah, 2019; Programme, 2024; WRAP, 2024b). While the IBWR system shows high potential to reduce emissions in the UK, applying it globally requires caution due to differences in waste composition, infrastructure, and policy capacity. Nonetheless, the UK result provides clear evidence of feasibility: with proper adaptation, similar integrated recovery systems could support emission reduction and economic gains worldwide, especially in regions with high organic waste generation. The IBWR model's main global value lies in offering a practical, scalable approach for sustainable FW management and resource recovery.

4.2. Policy recommendations

This study provides evidence-based policy insights showing that integrating BW valorisation with FW management through an optimization-based biorefinery system can simultaneously improve environmental, economic, and social outcomes. The analysis demonstrates that producing high-value biochemicals from the carbohydrate-, lipid-, and protein-rich fractions of BW enhances profitability, reduces greenhouse gas emissions, and supports green job creation. Incorporating carbon capture further strengthens climate benefits, while the heat supply strategy reveals a clear trade-off between emission reductions and economic gains. Based on this evidence, the following policy actions are recommended:

- **Promote integrated biorefineries producing high-value products.**
  Support the development of integrated systems that combine BW valorisation with existing FW management. Optimizing resource recovery and producing marketable products while enhancing environmental, economic, and social performance.
- **Focus on high-yield and high-value product pathways.**

Prioritize technological and process innovations that maximize conversion efficiency and target products with high market value derived from the carbohydrate, lipid, and protein fractions of BW. Focusing on high-yield, high-value outputs increases profitability, reduces emissions, and supports job creation.

- **Integrate carbon capture units in biorefinery systems.**
  Require the installation of carbon capture technologies to capture CO_2 from flue gas and off-gas streams, effectively reducing greenhouse gas emissions from waste processing facilities.
- **Adopt balanced heat supply strategies.**
  Recognize the trade-off between environmental and economic outcomes in heat supply. Supplying heat internally can lower emissions, while external heat supply increases revenue and job creation. Policies should support flexible strategies that balance both objectives.

4.3. Limitations and future works

Despite the novelty and innovative contribution of this study to FW management, several limitation remain, which also highlight important directions for future research. BW is generated across multiple stages of the food supply chain, with varying quantities and spatial distributions, making its collection a significant challenge (Narisetty et al., 2021). Although BW can be considered a relatively homogenous feedstock and is readily available from sources such as bakeries, retailers, and bread manufacturing facilities, its decentralized generation necessitates a coordinated collection strategy (Kumar et al., 2023). In the present study, BW collection is implicitly embedded within the feedstock cost, which allows the model to capture the influence of collection costs at an aggregate level. However, this approach does not explicitly account for the spatial distribution of waste sources, transportation routes, emissions associated with collection activities, or potential employment generated through collection and logistics. Future research could address this limitation by integrating a supply chain optimization framework into the IBWR system. Such a framework would enable a holistic assessment of optimal facility locations, transportation pathways, and collection strategies, thereby enhancing feedstock supply stability while reducing economic and environmental burdens. These problems can be formulated using MILP of location-routing models, or geographic information system (GIS)-based approaches, which have been widely in biomass supply chain studies (Bhatt et al., 2025; Moretti et al., 2021; Razm et al., 2019).

In addition to logistical challenges, uncertainty plays a critical role in the practical implementation of BW valorisation systems. In reality, BW characteristics fluctuate considerably across regions and sources due to differences in household consumption behaviour, dietary patterns and seasonal factors (Han et al., 2018). These fluctuations extend beyond quantity to include compositional variability in macronutrients such as carbohydrates, proteins and lipids, which directly influence conversion efficiency and process stability . Without explicitly accounting for such uncertainties, BW valorisation model may experience operational inefficiencies, economic variability, and unintended environmental trade-offs. Future studies could incorporate uncertainty analysis using Monte Carlo simulation (MCS), which is particularly effective for sampling from probability distributions to generate probabilistic outcomes. This approach allows for the identification of influential parameters, quantification of risk ranges, and evaluation of high-probability operational scenarios, thereby supporting more resilient decision-making (Dheskali et al., 2020). Alternatively, a two-stage stochastic optimization can be used to optimize decisions under uncertainty, capturing variations in process performance over time (Aranguren et al., 2021), whereas a robust optimization explicitly safeguards decisions against worst-case uncertainty scenarios (Huang et al., 2025).

Another important challenge in BW valorisation is managing variability in feedstock quality. This variability arises from both temporal and spatial differences in BW composition. For instance, BW collected from different geographical locations may exhibit distinct chemical profiles, while storage

duration and conditions can further alter moisture content and nutrient composition . Such variability complicates risk management, as fluctuations in feedstock quality can negatively affect conversion efficiency and introduce uncertainty into process economics (Ou et al., 2018). To mitigate these effects, future work could explore blending optimization strategies, in which BW is combined with other FW streams to achieve a more consistent and desirable feedstock composition. Blending optimization has been shown to reduce feedstock variability and improve process robustness in biomass and waste valorisation systems (Bhattarai et al., 2023; Ekşioğlu et al., 2021; Roni et al., 2018), making it a promising direction for extension in the current study.

5. Conclusion

This study presents the IBWR system as a novel and promising strategy for FW management, with a particular focus on valorizing BW. The framework is supported by a mathematical optimization model developed using the ε-constrained method in Pyomo, designed to identify the most profitable and sustainable process configurations within a biorefinery context. Case studies based on different heat supply strategies highlight clear trade-offs among economic, environmental, and social objectives. Single-objective optimization shows that Case 3, utilizing external heat sources, maximizes economic returns and employment (NPV: 733 M$; 4,881 jobs), processing 800 MT/day of BW to produce 291.17 MT/day of LA. In contrast, Case 1, relying solely on internal energy, achieves the highest environmental performance with 1,086 tCO_2-eq/day in GWP savings, processing 800 MT/day of BW with 76% directed to the biorefinery and 24% to incineration, yielding 220.57 MT/day of LA. Multi-objective optimization further reveals a spectrum of Pareto-optimal configurations that balance environmental, economic, and social goals. Economically favorable configurations tend to involve larger processing capacities, enhancing GWP savings and job creation. Notably, the 10th Pareto-optimal solution in Case 3 generates 733 M$ of NPV, reduces GWP by over 142,334 tCO_2-eq/year, and creates 3,187 jobs. Meanwhile, the highest GWP savings across multi-objective solutions is achieved in Case 1, reaching 361,741 tCO_2-eq/year. Overall, the IBWR framework provides a robust decision-support tool for designing integrated waste management systems. By reducing waste volumes and emissions, promoting resource recovery, and generating economic and social benefits, the IBWR system offers a scalable, sustainable pathway for circular economy implementation, contributing to environmental sustainability, economic resilience, and social well-being globally.

CRediT authorship contribution statement

Rendra Hakim Hafyan: Methodology, Resources, Formal analysis, Data curation, Visualization, writing – original draft, writing – review & editing. **Jhuma Sadhukhan:** Methodology, Writing – review & editing. **Siddharth Gadkari:** Funding acquisition, Methodology, Resources, Formal analysis, Data curation, Supervision, Visualization, writing – review & editing. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Akyuz, M. K., & Ekmekci, B. E. (2025). Environmental impacts of aviation-induced greenhouse gas emissions: a pre-to post COVID-19 analysis. *Aircraft Engineering and Aerospace Technology*, *97*(5), 566-577. https://doi.org/10.1108/AEAT-10-2024-0275

Aranguren, M., Castillo-Villar, K. K., & Aboytes-Ojeda, M. (2021). A two-stage stochastic model for co-firing biomass supply chain networks. *Journal of Cleaner Production*, *319*, 128582. https://doi.org/https://doi.org/10.1016/j.jclepro.2021.128582

Bacon, R., & Kojima, M. (2011). *Issues in estimating the employment generated by energy sector activities*. World Bank, Washingotn, DC.

Bakkaloglu, S., Cooper, J., & Hawkes, A. (2022). Life cycle environmental impact assessment of methane emissions from the biowaste management strategy of the United Kingdom: Towards net zero emissions. *Journal of Cleaner Production*, *376*, 134229. https://doi.org/https://doi.org/10.1016/j.jclepro.2022.134229

Ben Daya, B., & Nourelfath, M. (2019). Sustainability assessment of integrated forest biorefinery implemented in Canadian pulp and paper mills. *International Journal of Production Economics*, *214*, 248-265. https://doi.org/https://doi.org/10.1016/j.ijpe.2018.06.014

Bhatt, G., Upadhyay, A., & Sahoo, K. (2025). Biomass supply chain network design: Integrating fixed and portable preprocessing depots for cost efficiency and sustainability. *Applied Energy*, *389*, 125757. https://doi.org/https://doi.org/10.1016/j.apenergy.2025.125757

Bhattarai, A., Kemp, A., Jahromi, H., Kafle, S., & Adhikari, S. (2023). Thermochemical Characterization and Kinetics of Biomass, Municipal Plastic Waste, and Coal Blends and Their Potential for Energy Generation via Gasification. *ACS Omega*, *8*(48), 45985-46001. https://doi.org/10.1021/acsomega.3c06849

Brandão, A. S., Gonçalves, A., & Santos, J. M. R. C. A. (2021). Circular bioeconomy strategies: From scientific research to commercially viable products. *Journal of Cleaner Production*, *295*, 126407. https://doi.org/https://doi.org/10.1016/j.jclepro.2021.126407

Cambero, C., & Sowlati, T. (2016). Incorporating social benefits in multi-objective optimization of forest-based bioenergy and biofuel supply chains. *Applied Energy*, *178*, 721-735. https://doi.org/https://doi.org/10.1016/j.apenergy.2016.06.079

Chazara, P., Negny, S., & Montastruc, L. (2017). Quantitative method to assess the number of jobs created by production systems: Application to multi-criteria decision analysis for sustainable biomass supply chain. *Sustainable Production and Consumption*, *12*, 134-154. https://doi.org/https://doi.org/10.1016/j.spc.2017.07.002

Dheskali, E., Koutinas, A. A., & Kookos, I. K. (2020). Risk assessment modeling of bio-based chemicals economics based on Monte-Carlo simulations. *Chemical Engineering Research and Design*, *163*, 273-280. https://doi.org/https://doi.org/10.1016/j.cherd.2020.09.011

Dickson, R., Mancini, E., Garg, N., Woodley, J. M., Gernaey, K. V., Pinelo, M., Liu, J., & Mansouri, S. S. (2021). Sustainable bio-succinic acid production: superstructure optimization, techno-economic, and lifecycle assessment [10.1039/D0EE03545A]. *Energy & Environmental Science*, *14*(6), 3542-3558. https://doi.org/10.1039/D0EE03545A

Ekşioğlu, S. D., Gulcan, B., Roni, M., & Mason, S. (2021). A stochastic biomass blending problem in decentralized supply chains. *Naval Research Logistics (NRL)*, *68*(4), 434-453. https://doi.org/https://doi.org/10.1002/nav.21971

Facchini, E., Iacovidou, E., Gronow, J., & Voulvoulis, N. (2018). Food flows in the United Kingdom: The potential of surplus food redistribution to reduce waste. *Journal of the Air & Waste Management Association*, *68*(9), 887-899. https://doi.org/10.1080/10962247.2017.1405854

Frank, R. R., Cipullo, S., Garcia, J., Davies, S., Wagland, S. T., Villa, R., Trois, C., & Coulon, F. (2017). Compositional and physicochemical changes in waste materials and biogas production across 7 landfill sites in UK. *Waste Management*, *63*, 11-17. https://doi.org/https://doi.org/10.1016/j.wasman.2016.08.026

Gadkari, S., Kumar, D., Qin, Z.-h., Ki Lin, C. S., & Kumar, V. (2021). Life cycle analysis of fermentative production of succinic acid from bread waste. *Waste Management*, *126*, 861-871. https://doi.org/https://doi.org/10.1016/j.wasman.2021.04.013

Galanopoulos, C., Kenkel, P., & Zondervan, E. (2019). Superstructure optimization of an integrated algae biorefinery. *Computers & Chemical Engineering*, *130*, 106530. https://doi.org/https://doi.org/10.1016/j.compchemeng.2019.106530

Giuliano, A., Poletto, M., & Barletta, D. (2016). Process optimization of a multi-product biorefinery: The effect of biomass seasonality. *Chemical Engineering Research and Design*, *107*, 236-252. https://doi.org/https://doi.org/10.1016/j.cherd.2015.12.011

GOV.UK. (2024). *Green Gas Support Scheme (GGSS): Annual Tariff Review 2024*. Retrieved 10 November from https://www.gov.uk/government/publications/green-gas-support-scheme-ggss-annual-tariff-reviews-and-tariff-change-notices/green-gas-support-scheme-ggss-annual-tariff-review-2024

Hafyan, R. H., Bhullar, L. K., Mahadzir, S., Bilad, M. R., Nordin, N. A., Wirzal, M. D., Putra, Z. A., Rangaiah, G. P., & Abdullah, B. (2020). Integrated Biorefinery of Empty Fruit Bunch from Palm Oil Industries to Produce Valuable Biochemicals. *Processes*, *8*(7).

Hafyan, R. H., Mohanarajan, J., Uppal, M., Kumar, V., Narisetty, V., Maity, S. K., Sadhukhan, J., & Gadkari, S. (2024a). Bread waste valorization: a review of sustainability aspects and challenges [Review]. *Frontiers in Sustainable Food Systems*, *8*. https://doi.org/10.3389/fsufs.2024.1334801

Hafyan, R. H., Mohanarajan, J., Uppal, M., Kumar, V., Narisetty, V., Maity, S. K., Sadhukhan, J., & Gadkari, S. (2024b). Integrated biorefinery for bioethanol and succinic acid co-production from bread waste: Techno-economic feasibility and life cycle assessment. *Energy Conversion and Management*, *301*, 118033. https://doi.org/https://doi.org/10.1016/j.enconman.2023.118033

Hafyan, R. H., Sadhukhan, J., Kumar, V., Maity, S. K., & Gadkari, S. (2025). A comparative techno-economic feasibility of hydrogen production from sugarcane bagasse and bread waste. *Fuel*, *388*, 134469. https://doi.org/https://doi.org/10.1016/j.fuel.2025.134469

Haghpanah, T., Sobati, M. A., & Pishvaee, M. S. (2022). Multi-objective superstructure optimization of a microalgae biorefinery considering economic and environmental aspects. *Computers & Chemical Engineering*, *164*, 107894. https://doi.org/https://doi.org/10.1016/j.compchemeng.2022.107894

Han, Z., Liu, Y., Zhong, M., Shi, G., Li, Q., Zeng, D., Zhang, Y., Fei, Y., & Xie, Y. (2018). Influencing factors of domestic waste characteristics in rural areas of developing countries. *Waste Management*, *72*, 45-54. https://doi.org/https://doi.org/10.1016/j.wasman.2017.11.039

Hanna, R., Heptonstall, P., & Gross, R. (2024). Job creation in a low carbon transition to renewables and energy efficiency: a review of international evidence. *Sustainability Science*, *19*(1), 125-150. https://doi.org/10.1007/s11625-023-01440-y

Hannah, R. (2019). Food production is responsible for one-quarter of the world's greenhouse gas emissions. *Our World in Data*. https://ourworldindata.org/food-ghg-emissions

Hong, W. Y. (2022). A techno-economic review on carbon capture, utilisation and storage systems for achieving a net-zero CO2 emissions future. *Carbon Capture Science & Technology*, *3*, 100044. https://doi.org/https://doi.org/10.1016/j.ccst.2022.100044

Huang, X., Ji, L., Xie, Y., & Luo, Z. (2025). Robust optimization of regional biomass supply chain system design and operation with data-driven uncertainties. *Food and Bioproducts Processing*, *149*, 176-189. https://doi.org/https://doi.org/10.1016/j.fbp.2024.11.021

Jeswani, H. K., Figueroa-Torres, G., & Azapagic, A. (2021). The extent of food waste generation in the UK and its environmental impacts. *Sustainable Production and Consumption*, *26*, 532-547. https://doi.org/https://doi.org/10.1016/j.spc.2020.12.021

Karmee, S. K., Linardi, D., Lee, J., & Lin, C. S. K. (2015). Conversion of lipid from food waste to biodiesel. *Waste Management*, *41*, 169-173. https://doi.org/https://doi.org/10.1016/j.wasman.2015.03.025

Kumar, V., Brancoli, P., Narisetty, V., Wallace, S., Charalampopoulos, D., Kumar Dubey, B., Kumar, G., Bhatnagar, A., Kant Bhatia, S., & Taherzadeh, M. J. (2023). Bread waste – A potential feedstock for sustainable circular biorefineries. *Bioresource Technology*, *369*, 128449. https://doi.org/https://doi.org/10.1016/j.biortech.2022.128449

Mailaram, S., Narisetty, V., Maity, S. K., Gadkari, S., Thakur, V. K., Russell, S., & Kumar, V. (2023). Lactic acid and biomethane production from bread waste: a techno-economic and profitability analysis using pinch technology [10.1039/D3SE00119A]. *Sustainable Energy & Fuels*, *7*(13), 3034-3046. https://doi.org/10.1039/D3SE00119A

Maity, S. K. (2015). Opportunities, recent trends and challenges of integrated biorefinery: Part II. *Renewable and Sustainable Energy Reviews*, *43*, 1446-1466. https://doi.org/https://doi.org/10.1016/j.rser.2014.08.075

Manhongo, T. T., Chimphango, A., Thornley, P., & Röder, M. (2025). A socio-economic and multi-criterion decision analysis approach for a holistic sustainability assessment of mango waste-based integrated biorefineries. *Environment, Development and Sustainability*. https://doi.org/10.1007/s10668-025-06851-8

Moretti, L., Milani, M., Lozza, G. G., & Manzolini, G. (2021). A detailed MILP formulation for the optimal design of advanced biofuel supply chains. *Renewable Energy*, *171*, 159-175. https://doi.org/https://doi.org/10.1016/j.renene.2021.02.043

Narisetty, V., Cox, R., Willoughby, N., Aktas, E., Tiwari, B., Matharu, A. S., Salonitis, K., & Kumar, V. (2021). Recycling bread waste into chemical building blocks using a circular biorefining approach [10.1039/D1SE00575H]. *Sustainable Energy & Fuels*, *5*(19), 4842-4849. https://doi.org/10.1039/D1SE00575H

Narisetty, V., Nagarajan, S., Gadkari, S., Ranade, V. V., Zhang, J., Patchigolla, K., Bhatnagar, A., Kumar Awasthi, M., Pandey, A., & Kumar, V. (2022). Process optimization for recycling of bread waste into bioethanol and biomethane: A circular economy approach. *Energy Conversion and Management*, *266*, 115784. https://doi.org/https://doi.org/10.1016/j.enconman.2022.115784

Nikolaou, M., Stavraki, C., Bousoulas, I., Malamis, D., Loizidou, M., Mai, S., & Barampouti, E. M. (2023). Valorisation of bakery waste via the bioethanol pathway. *Energy*, *280*, 128185. https://doi.org/https://doi.org/10.1016/j.energy.2023.128185

Ou, L., Luo, G., Ray, A., Li, C., Hu, H., Kelley, S., & Park, S. (2018). Understanding the Impacts of Biomass Blending on the Uncertainty of Hydrolyzed Sugar Yield from a Stochastic Perspective. *ACS Sustainable Chemistry & Engineering*, *6*(8), 10851-10860. https://doi.org/10.1021/acssuschemeng.8b02150

Ouro-Salim, O., & Guarnieri, P. (2022). Circular economy of food waste: A literature review. *Environmental Quality Management*, *32*(2), 225-242. https://doi.org/https://doi.org/10.1002/tqem.21836

Park, S., Mun, H., Park, J., & Lee, I. (2024). Cost optimization methodology based on carbon-techno-economic analysis: An application to post-combustion carbon capture process. *Journal of Cleaner Production*, *434*, 139887. https://doi.org/https://doi.org/10.1016/j.jclepro.2023.139887

Programme, U. N. E. (2024). *Food Waste Index Report 2024. Think Eat Save: Tracking Progress to Halve Global Food Waste.* https://wedocs.unep.org/20.500.11822/45230.

Razm, S., Nickel, S., & Sahebi, H. (2019). A multi-objective mathematical model to redesign of global sustainable bioenergy supply network. *Computers & Chemical Engineering*, *128*, 1-20. https://doi.org/https://doi.org/10.1016/j.compchemeng.2019.05.032

Roni, M. S., Thompson, D., Hartley, D., Searcy, E., & Nguyen, Q. (2018). Optimal blending management of biomass resources used for biochemical conversion. *Biofuels, Bioproducts and Biorefining*, *12*(4), 624-648. https://doi.org/https://doi.org/10.1002/bbb.1877

Sadhukhan, J., Gadkari, S., Martinez-Hernandez, E., Ng, K. S., Shemfe, M., Torres-Garcia, E., & Lynch, J. (2019). Novel macroalgae (seaweed) biorefinery systems for integrated chemical, protein, salt, nutrient and mineral extractions and environmental protection by green synthesis and life cycle sustainability assessments [10.1039/C9GC00607A]. *Green Chemistry*, *21*(10), 2635-2655. https://doi.org/10.1039/C9GC00607A

Sadhukhan, J., & Martinez-Hernandez, E. (2017). Material flow and sustainability analyses of biorefining of municipal solid waste. *Bioresource Technology*, *243*, 135-146. https://doi.org/https://doi.org/10.1016/j.biortech.2017.06.078

Sadhukhan, J., Martinez-Hernandez, E., Amezcua Allieri, M. A., Zermeño Eguía-Lis, J. A., Castillo, A., Dominguillo, D., Torres-García, E., & Aburto, J. (2024). Strategic navigation of world-leading biorefineries and Mexico's policy landscape: A gateway to a sustainable circular bioeconomy. *Journal of Cleaner Production*, *434*, 140386. https://doi.org/https://doi.org/10.1016/j.jclepro.2023.140386

Sadhukhan, J., Ng, K. S., & Hernandez, E. M. (2014). *Biorefineries and chemical processes: design, integration and sustainability analysis*. John Wiley & Sons.

Sadhukhan, J., Ng, K. S., & Martinez-Hernandez, E. (2016). Novel integrated mechanical biological chemical treatment (MBCT) systems for the production of levulinic acid from fraction of municipal solid waste: A comprehensive techno-economic analysis. *Bioresource Technology*, *215*, 131-143. https://doi.org/https://doi.org/10.1016/j.biortech.2016.04.030

Sadhukhan, J., Sen, S., & Gadkari, S. (2021). The Mathematics of life cycle sustainability assessment. *Journal of Cleaner Production*, *309*, 127457. https://doi.org/https://doi.org/10.1016/j.jclepro.2021.127457

Sinha, S., & Tripathi, P. (2021). Trends and challenges in valorisation of food waste in developing economies: A case study of India. *Case Studies in Chemical and Environmental Engineering*, *4*, 100162. https://doi.org/https://doi.org/10.1016/j.cscee.2021.100162

Slorach, P. C., Jeswani, H. K., Cuéllar-Franca, R., & Azapagic, A. (2020). Assessing the economic and environmental sustainability of household food waste management in the UK: Current situation and future scenarios. *Science of The Total Environment*, *710*, 135580. https://doi.org/https://doi.org/10.1016/j.scitotenv.2019.135580

Svensson, S. E., Bucuricova, L., Ferreira, J. A., Souza Filho, P. F., Taherzadeh, M. J., & Zamani, A. (2021). Valorization of Bread Waste to a Fiber- and Protein-Rich Fungal Biomass. *Fermentation*, *7*(2).

Thomassen, G., Van Dael, M., You, F., & Van Passel, S. (2019). A multi-objective optimization-extended techno-economic assessment: exploring the optimal microalgal-based value chain [10.1039/C9GC03071A]. *Green Chemistry*, *21*(21), 5945-5959. https://doi.org/10.1039/C9GC03071A

Uebbing, J., Rihko-Struckmann, L., Sager, S., & Sundmacher, K. (2020). CO2 methanation process synthesis by superstructure optimization. *Journal of CO2 Utilization*, *40*, 101228. https://doi.org/https://doi.org/10.1016/j.jcou.2020.101228

Wang, D., He, J., Tang, Y.-T., Higgitt, D., & Robinson, D. (2020). Life cycle assessment of municipal solid waste management in Nottingham, England: Past and future perspectives. *Journal of Cleaner Production*, *251*, 119636. https://doi.org/https://doi.org/10.1016/j.jclepro.2019.119636

WPI. (2022). *Green Jobs and Skills in London: cross-London report*

WRAP. (2023). *Food Surplus and Waste in the UK - Key Facts November 2023.* . https://www.wrap.ngo/resources/report/food-surplus-and-waste-uk-key-facts-updated-november-2023#download-file

WRAP. (2024a). *Gate Fees Report 2023/24: Comparing the costs of alternative waste treatment options]*.

WRAP. (2024b). Global Action Drive For Food Loss and Waste Reduction https://www.wrap.ngo/resources/report/global-action-drive-food-loss-and-waste-reduction

Wu, Y., Restrepo-Flórez, J. M., & Vasco-Correa, J. (2024). A Superstructure-Based Lignin Valorization Process Optimization Model for Lignocellulosic Biorefineries through Biological Upgrading. *ACS Sustainable Chemistry & Engineering*, *12*(18), 6913-6926.

Yaashikaa, P. R., Kumar, P. S., Saravanan, A., Varjani, S., & Ramamurthy, R. (2020). Bioconversion of municipal solid waste into bio-based products: A review on valorisation and sustainable approach for circular bioeconomy. *Science of The Total Environment*, *748*, 141312. https://doi.org/https://doi.org/10.1016/j.scitotenv.2020.141312